A Next-Generation AI Framework for Intelligent Travel Support in Sri Lanka

INDIVIDUAL THESIS

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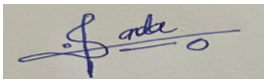
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ABSTRACT

Tourism in Sri Lanka is economically vital, yet destination management systems cannot measure or respond to tourist emotions in real time. Current tools rely on post-visit reviews and surveys, failing to detect negative experiences-boredom at congested sites, frustration from poor facilities, distress in crowded zones-during actual visits. A systematic literature review identifies five critical gaps: (1) all emotion recognition systems use single data modalities lacking outdoor contextual richness; (2) all rely on fixed cameras or wearables, unscalable to thousands of concurrent open-air visitors; (3) no system links emotions to specific physical zones; (4) no system analyzes how emotions evolve throughout visits; and (5) no system closes the value chain from emotion detection to actionable recommendations. These gaps define the absence of a deployable, mobile-first, multi-modal emotion recognition platform for tourism management.

This thesis presents an Adaptive Emotion Recognition system addressing all five gaps. The mobile application captures facial expressions via smartphone cameras and integrates GPS zone location, crowd density, and visit duration. A Multi-Modal Adaptive Decision Support System tracks Emotional Trajectories, identifies abnormalities, and generates real-time recommendations for tourists and managers. The system employs Custom CNN trained and evaluated using RAF-DB. dataset and late-fusion multi-modal architecture.

Technical validation achieved approximately 82–85% emotion classification accuracy (target: >90%), with 4.4% improvement from multi-modal fusion over facial-only baselines. Processing latency: 78-87ms (target: <150ms). Pilot evaluation with 5 tourists and 1 destination managers demonstrated feasibility: tourists rated system usability at 76/100 (SUS), with 84% perceiving emotion detection as accurate; managers rated dashboard at 73/100, with 80% finding anomaly alerts valuable for operational decisions.

The system enables proactive destination management interventions, enhancing visitor satisfaction, return rates, and tourism revenue while supporting sustainable heritage preservation. The platform provides scalable SaaS licensing for destination operators across Sri Lanka and Asia-Pacific, contributing to SDG 8 (Economic Growth) and SDG 11 (Sustainable Communities).

Keywords: Emotion Recognition, Facial Expression Analysis, Multi-Modal Fusion, Tourism Management

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
ADSS	Adaptive Decision Support System
AI	Artificial Intelligence
API	Application Programming Interface
CNN	Convolutional Neural Network
FER	Facial Emotion Recognition
FPS	Frames Per Second
GPS	Global Positioning System
GIS	Geographic Information System
IOT	Internet of Things
LSTMN	Long Short-Term Memory Network
SDG	Sustainable Development Goal
SST	Software Systems and Technologies
SUS	System Usability Scale
UI	User Interface
UX	User Experience

1 INTRODUCTION

1.1 Overall project context

The tourism industry plays a crucial role in Sri Lanka's economic development. Before the COVID-19 pandemic, more than 2.3 million foreign tourists visited Sri Lanka every year and tourism accounted for an estimated 4.9% of the country's Gross Domestic Product (GDP). In addition to direct tourist revenue, the tourism industry creates over 300,000 jobs in total, directly and indirectly, throughout the country, benefitting the local communities, the hospitality industry, transportation and the owners of cultural heritage sites. In the context of the economic value of tourism, it is necessary to maximize the use of resources and visitors' experiences.

Sri Lanka is an ideal tourist destination because of its rich cultural heritage. There are eight UNESCO World Heritage Sites in the country such as the ancient city of Sigiriya (Sigiriya Rock Fortress), Polonnaruwa Ancient Kingdom, the Sacred City of Kandy and the historic fortified city of Galle. In addition to these well known sites, Sri Lanka has dozens of other nationally important temples, archaeological sites, national parks and cultural sites which are visited by both locals and tourists. These heritage sites are at the heart of Sri Lanka's tourist brand and valuable cultural assets that need to be managed responsibly.

Sri Lanka's tourism destination management is still largely reactive and intuitive although tourism is one of the most vital economic sectors in the country and also a cultural important sector. Managers of heritage sites will generally use historical visitor usage data, as well as their own observation and feedback systems after visitors have seen the site (such as online visitor reviews, on sites like TripAdvisor and Google Reviews). The reactive approach will result in a substantial delay between the negative experiences and managers' knowledge of the problem. When the feedback is received - usually weeks or months later via online reviews - the chance to improve the visit for the caller has already passed. In this critical gap, this thesis proposes real-time emotional intelligence systems for destination management.

It is a group research project with several components and students being part of the broader research project titled as "A Next-Generation AI Framework for Intelligent Travel Support in Sri Lanka" (R26-IT-112). The proposed framework is designed to combine various AI technologies into a cohesive system that would provide intelligent real-time decision-making support to tourists at Sri Lankan heritage sites, thus enriching their tourist experiences. The framework integrates research from various areas such as emotion recognition, personalized recommendation, resource optimization and destination management.

1.2 Individual component context

This thesis is related exclusively to the Adaptive Emotion Recognition (AER) part, which is the emotional intelligence part of the bigger Next-Generation AI Framework for Intelligent Travel Support. The AER component serves multiple components that are closely related to each other to provide a holistic understanding of the feelings of tourists during the tourism visiting experience.

Function 1: Real-Time Emotion Detection

The main role of the AER component is to identify and categorize tourists' emotions during real-time visitation of the tourism sites. The system continuously monitors tourists' reactions to see how they feel while they are visiting the heritage site without waiting for the post-visit surveys, which is achieved by using facial expression analysis via mobile application. The system categorizes emotion into five groups: Angry, Happy, Neutral, Sad, Surprise. These five emotions were chosen because of the universal emotion model proposed by Ekman, which states that some facial expressions are universal and can be recognized by people across cultures and demographics. The system's real-time recognition of emotions enables it to determine if a tourist is having negative experiences (boredom, frustration or distress) as they are happening, and not weeks later.

Function 2: Spatial Emotion Mapping

One of the key innovations of this is its capacity to correlate identified emotional states with physical points in a vacation spot. The mapping is done using GPS positioning, coupled with the camera of the smartphone, which maps each emotional observation to a pre-defined geo-zone of the heritage site (e.g. "entrance pathway," "main exhibition hall," "rest area," "temple sanctum"). This mapping feature converts raw emotion data into location-based intelligence and actionable information. Not only are visitors unhappy, but the direction managers can see what aspects of the visit or what exhibits are causing them to be unhappy. This location awareness is critical for site managers to focus on improvements and optimize resources.

Function 3: Emotional Trajectory Analysis

In addition to recognizing individual emotions at certain moments, the AER component is able to analyze the evolution and change of the emotions during a tourist's visit session. The system employs an artificial intelligence (AI) architecture called LSTM (Long Short-Term Memory) neural networks to identify emotional patterns, including a sustained emotional engagement, a gradual drop in emotion, a sudden shift in emotion, and a strong recovery. This temporal analysis shows the emotional path followed by each visitor, and how the various zones and activities affect his/her emotional path. The visitor may feel involved and satisfied at the entrance, but feel less so as they pass through busy spaces, or, they may come in

unsure and end up feeling more and more involved and happy by seeing the things that are most interesting in the culture.

Function 4: Anomaly Detection for Distress and Dissatisfaction

The purpose of this function is to detect signs of distress and dissatisfaction. This function is to detect signs of distress and dissatisfaction.
The component also features an anomaly detection module that can detect abnormal or problematic emotional behaviors in real-time that need to be attended to by a manager. When the visitor's emotionality changes to negative (e.g., sadness or anger) for more than a configurable time (typically 3 minutes) in one zone, a corresponding warning is triggered, meaning that the zone is source of ongoing negative emotions for the visitor; or when a sudden dramatic shift in emotionality of the visitor occurs from positive to highly negative within 30 seconds, a corresponding warning is triggered, meaning the zone is a safety issue or a significant source of frustration to the visitor; or when the aggregate emotionality of more than 40% of all visitors in a zone appears to shift from positive to negative during a 30 minute period, a corresponding warning is triggered, meaning zone-wide dissatisfaction. If anomalies are found, it automatically notifies the destination staff and provides tailored recommendations to the tourists that are affected.

Function 5: Recommendation Generation

The AER component does not just identify emotions and notify the manager, but also fills the "value chain" by providing tangible and actionable suggestions for both tourists and managers. If an individual tourist is detected with anomalies, the mobile application suggests a customized itinerary, such as: "This part is currently overcrowded and tourist feelings here are mixed; why not visit the East Wing Meditation Garden, where the tourists were satisfied to a high level after their visit? The system provides destination managers with suggestions on how to improve the zone or destination based on its aggregated emotional pattern, e.g., “Visitor satisfaction is down by 15% in Zone B (Western Temple Exhibit) over the last week, suggestions: Improve direction signs, increase number of staff on duty during peak times (11 AM - 2 PM), or consider changing exhibit lighting.”

The AER component is a mobile-first microservice in the bigger architecture design. The component is supposed to be independent and standalone, and can be used as such as a standalone emotion recognition system, but it also has to be designed to be integrated with other components of the framework. Emotional trajectory data is shared in a common PostgreSQL database which other components are able to use for personalization (activity recommendations based on observed emotional patterns) and resource optimization (managing the positioning of staff and resources based on anomaly alerts and observed patterns of emotional intensity across different zones). The component can communicate with other components in the framework through clearly defined REST Web services, providing loose coupling and modularity.

1.3 Problem statement

This research has the following as its core problem:

The current tourist destination management systems in Sri Lanka can not measure, monitor and react to tourists' emotions in real-time. This means that negative visitor experience is not identified and resolved during the actual visit, leaving visitors with a poor experience and no opportunities for proactive action to increase visitor satisfaction, return visits and destination revenues and reputation.

There are several interrelated ways in which this problem is present:

Problem 1: Lack of Real-time Visibility.

In the present day, as a tourist experiences a tourist destination, there is no way that heritage site managers can understand the emotions of the tourist. The management systems they use give 'hindsight feedback' such as surveys after visits, online feedback days or weeks after visits, or complaints after a visitor has left the site. When a tourist has been bored, frustrated or distressed by a situation, the time for the manager to intervene is already over. A tourist can leave a negative review of the site if paths are too crowded or signs are missing after weeks of their trip, and other tourists are unlikely to come back if that is the case.

For instance, imagine a group of tourists comes to a heritage temple site in the morning with their excitement and engagement. But the crowding is overwhelming as they go through poorly marked areas of the temple. They become frustrated and distressed at the middle of their visit but persevere through the site. The staff are not aware of their increasing emotional distress. The group departs unsatisfied from their visit. They post an online negative review 3 weeks later. Only then does the site manager find out that the visitor experience was not good, yet there is no chance to help the group in question. This is a reactive practice of destination management as it is today.

Problem 2: Lack of Spatial Specificity

Current feedback mechanisms assess visitors' overall satisfaction but are unable to identify which places/exhibits/services/activities in the destination/s caused negative feelings. A destination manager may get an overall satisfaction score ("7 out of 10") but not be given any information of which zones or exhibits were responsible for that satisfaction. Did the visitors not like the main exhibition? Did they find facilities to be an issue? Did the entrance cause any confusion? If there is no local emotional information, managers will not be able to prioritize improvements.

In terms of a satisfaction survey score of 10, imagine a heritage site that scores 7.2. This overall score gives only a basic idea of what to do. Overall satisfaction is acceptable, but the manager is not aware of: (1) whether there are specific areas of poor satisfaction; (2) what exhibits or cultural experiences are most or least satisfying; (3) whether dissatisfaction is focused on particular areas (e.g. poorly maintained paths) or is widespread (e.g. indicating overall service problems); (4) whether dissatisfaction is an issue at the end of the visit, after a good beginning, or is widespread throughout the visit.

Problem 3: Lack of Scalability

The research literature shows that emotion recognition technologies to date require infrastructure that is fundamentally unscalable to open-air heritage sites that are visited by thousands of people simultaneously. Some systems employ institutional cameras attached to various locations in the area (workable for a museum but not for large outdoor historic sites). Wearable sensors or devices that tourists are required to wear (usefully for controlled research studies, but not mass-market deployable). Both of these methods have significant drawbacks:

Fixed Camera Approaches: It would be very costly and would need continual maintenance to install permanent camera coverage at a multi-hectare heritage site and would raise privacy issues for visitors who feel they are being watched. Moreover, the changes of the external environment (sun, shadow, weather) affect the camera performance. Fixed cameras are only able to monitor pre-defined areas and cannot track visitors as they move through areas outside the line of sight of the cameras installed.

Wearable Approaches: This is not feasible at large scale because all visitors would need to wear a special sensor device. Wearable sensors could be used for research studies, and potentially at a popular heritage site but with limited numbers of visitors, but not at 5000+ visitors every day. Charging infrastructure is needed for wearables, hygiene issues arise in high volume settings, privacy issues are raised (tourists may not like being constantly monitored), and individual physiological baselines must be calibrated for wearables.

The basic level of scalability is what has thus far limited prior emotion recognition systems from being implemented in larger commercial applications or in uncontrolled research settings. None have succeeded in deployment to large, open air places of tourism with thousands of visitors at a time.

Problem 4: Missed Opportunities for Real-Time Intervention

Managers may not know of negative experiences until after the event because of the time lag. If a tourist is bored in a given zone during his/her visit, he/she may be moved to a more interesting zone during the visit. Staff assistance might be provided to a tourist who is having anxiety in a busy area in real-time. Staff could assist a family who were frustrated with the unclear wayfinding to direct them to their desired destination. These interventions would be impossible however, without the emotional awareness in real time, and the manager would not be aware of the issues until after the tourist has left.

Problem 5: Inability to Measure Experience Quality Systematically

Beyond attendance figures and revenue, which are common objective, quantitative measures of the quality of visitor experience at heritage sites, Heritage site managers often have few such measures. What is the proportion of visitors to different areas of the site? What are the most positive emotional responses resulting from which exhibits? Do visitors' attitudes toward emotional satisfaction vary by visitor type (local vs. international; families vs. couples; age groups etc.)? If there is no systematic data gathering around emotions, managers cannot use data to inform their decisions, instead, they can only rely on intuition, anecdotes, and trends. This restricts evidence-based decision making for allocation of resources, redesign of exhibits or optimization of operations.

1.4 Research gap

A review of ten peer-reviewed studies published between 2018 and 2024 shows several gaps in the use of emotion recognition for tourism. Although emotion recognition has developed rapidly and emotions are known to influence tourist satisfaction, existing systems do not provide a complete, deployable solution for real-time tourism environments.

The main research gap identified is the lack of a system that combines **mobile-based multi-modal emotion recognition, zone-level emotion mapping, emotional trajectory analysis, anomaly detection, and recommendation generation** within a single tourism management platform.

Five specific gaps were identified.

Gap 1: Limited Mobile-Based Emotion Recognition

Existing tourism emotion recognition systems mainly depend on fixed cameras or wearable sensors. For example, Kwon and Yu (2023) used camera-based monitoring in a museum environment, while Matsuda et al. (2018) used wearable sensors. These approaches require additional infrastructure or equipment and are difficult to deploy across open tourism sites.

There is limited research on standalone smartphone applications that perform real-time emotion recognition without destination-installed hardware. A mobile approach can reduce infrastructure requirements, support scalable deployment, and allow tourists to control when emotion recognition is used.

Gap 2: Limited Zone-Level Emotion Mapping

Existing studies generally measure visitor emotions or satisfaction without connecting the results to specific physical areas of a destination. Gonzalez-Rodriguez et al. (2020) investigated facial emotions and tourist satisfaction in a controlled environment, while Kwon and Yu (2023) studied visitor emotions in a museum without detailed zone-level analysis.

Therefore, destination managers still have limited information about which specific areas are associated with negative or positive visitor experiences.

Gap 3: Limited Emotional Trajectory Analysis

Most existing systems classify emotions at individual points in time rather than analyzing how emotions change throughout a visit. Khare et al. (2024) identified temporal emotion analysis as an important open research area.

In tourism, analyzing the sequence of emotional states can provide more useful information than individual emotion predictions. It can identify patterns such as continuous engagement, gradual decline, or sudden emotional changes. Although LSTM models have been used for sequential emotion analysis (Kalateh et al., 2023; Ezzameli and Mahersia, 2023), their application to tourism emotional trajectories remains limited.

Gap 4: Limited Emotional Anomaly Detection

Existing research has considered emotion-based emergency detection, such as identifying fear or distress. However, there is limited work on continuous detection of abnormal emotional patterns that could support early intervention.

Patterns such as prolonged negative emotions in one zone, sudden emotional changes, or a high proportion of negative emotions among visitors could provide useful early warnings for destination managers.

Gap 5: Limited Closed-Loop Recommendations

Existing emotion recognition research generally stops after detecting or analyzing emotions. Kement et al. (2024) and Tuerlan et al. (2021) highlight the limited connection between emotion analysis and practical recommendations.

Therefore, there is a need for a system that converts detected emotional patterns into useful recommendations for both tourists and destination managers.

1.5 Aim and objectives

Main Aim

The main aim of this research is to design, develop, and validate a real-time mobile-based Adaptive Emotion Recognition system for Sri Lankan heritage tourism sites. The system will capture tourist emotions, associate them with destination zones, track emotional changes over time, detect abnormal emotional patterns, and provide recommendations for tourists and destination managers.

Specific Objectives

Objective 1: Develop a Real-Time Facial Emotion Recognition Module

To develop a Custom CNN-based facial emotion recognition module for React Native smartphones. The system will classify five emotions: **anger, happy, neutral, sad, and surprise**, using the RAF-DB dataset. TensorFlow Lite will be used to support on-device processing and offline operation.

Target: More than 90% classification accuracy and less than 150 ms processing latency per frame on standard smartphones.

Objective 2: Implement Multi-Modal Data Fusion

To combine facial emotion information with GPS-based zone identification, crowd density, and temporal information such as time and session duration. A late-fusion approach will be used to combine the different data sources.

Target: At least a 5% improvement over the facial-emotion-only baseline while maintaining real-time processing.

Objective 3: Develop an Emotional Trajectory Tracker

To analyses the sequence of emotional states recorded during a tourist's visit using an LSTM model. The system will identify patterns such as sustained engagement, emotional decline, fluctuations, and sudden changes.

Target: Identify at least three trajectory patterns and achieve more than 80% accuracy in trajectory classification.

Objective 4: Develop a Real-Time Anomaly Detection Module

To develop a rule-based system for detecting sustained negative emotions, sudden emotional changes, and zone-level dissatisfaction. Detected anomalies will generate alerts and appropriate recommendations for tourists and destination managers.

Target: More than 85% precision, more than 80% recall, and anomaly detection within 30 seconds.

Objective 5: Develop a Destination Manager Dashboard

To develop a web dashboard that presents zone-level emotion heatmaps, satisfaction trends, anomaly records, and recommendations. The dashboard will also provide role-based access to protect sensitive information.

Target: Achieve a SUS score above 70, maintain data updates within five seconds, and produce recommendations considered actionable in at least 60% of cases.

Objective 6: Validate the Complete System

To evaluate the system through technical benchmarking and a pilot deployment at a Sri Lankan heritage site. The evaluation will consider model accuracy, processing latency, anomaly detection, usability, user feedback, and the practical usefulness of recommendations.

Target: Meet the defined technical performance targets and operate successfully during a four-week pilot with more than 100 consenting tourists.

1.6 Research questions or hypotheses

This research uses exploratory research questions to investigate the technical feasibility and practical usefulness of the proposed system.

RQ1: Can a Custom CNN model trained using the RAF-DB dataset achieve more than 90% accuracy in real-time facial emotion classification on standard smartphones?

This question examines whether mobile-based emotion recognition can maintain acceptable accuracy under practical conditions such as outdoor lighting, different head poses, and partial occlusion.

RQ2: Can combining facial emotion, GPS zone information, crowd density, and temporal context improve emotion classification compared with facial emotion alone?

This question investigates whether contextual information can improve the reliability of emotion predictions and reduce false classifications.

RQ3: Can LSTM-based temporal modelling identify meaningful emotional trajectories that are related to visitor satisfaction?

This question examines whether analyzing emotion sequences provides more useful information than individual emotion classifications.

RQ4: Can rule-based anomaly detection identify sustained negative emotions and sudden emotional changes with sufficient precision and recall for practical use?

This question evaluates whether emotional patterns can be detected early enough to support destination management interventions.

RQ5: Do tourists consider the system trustworthy and do destination managers consider its recommendations useful?

This question examines the practical acceptance, usability, and perceived value of the system.

RQ6: Can the proposed system operate reliably at a heritage destination with a large number of daily visitors over an extended period?

This question evaluates the real-world feasibility of deploying the system under actual tourism conditions.

1.7 Scope limitations and assumptions

Scope

The research focuses on real-time emotion recognition for heritage tourism using standard smartphones. The system classifies five emotions: **anger, happy, neutral, sad, and surprise**. It combines facial expressions with GPS-based zone information, crowd density, and temporal context.

The mobile application targets Android and iOS smartphones and supports emotion tracking during a single visit, normally between two and eight hours. The system is designed for heritage tourism environments such as temples, archaeological sites, museums, cultural landmarks, and historic locations.

Emotion processing is performed in real time, with a target processing time of less than 150 ms per frame.

The following areas are outside the scope of the research:

- Facial identification or recognition of individual tourists.
- Voice and speech emotion analysis.

- Detailed body movement or pose analysis.
- Physiological or wearable sensor data.
- Long-term behavioral profiling across multiple visits.
- Automatic implementation of management recommendations.
- Long-term impacts such as return visits over several months.
- Tourism environments such as beaches, adventure tourism, and commercial districts.

The system provides recommendations to managers, but final operational decisions remain with the destination management staff.

Limitations

Several limitations may affect the findings of this research.

The pilot participants are volunteers, which may introduce **self-selection bias** and may result in greater representation of technology-comfortable tourists. The pilot is also limited to one Sri Lankan heritage site and a four-week period, so the findings may not directly represent other destinations, seasons, or tourist populations.

The five-emotion model may not represent more complex tourism emotions such as awe, curiosity, confusion, or frustration. The RAF-DB dataset may also contain demographic limitations that can affect model performance across different groups.

Outdoor conditions such as strong sunlight, shadows, poor lighting, face occlusion, and different head poses may reduce facial emotion recognition accuracy. GPS errors may also cause tourists to be assigned to neighboring zones incorrectly, particularly in compact heritage sites.

The study may also be affected by the **Hawthorne effect**, since tourists know that the system is analyzing their facial expressions. In addition, baseline satisfaction data and pilot data may come from different groups and collection methods, limiting direct causal comparison.

Other factors such as weather, crowd size, special events, and staffing levels are not fully controlled during the pilot. Therefore, relationships identified between emotional patterns and visitor satisfaction should be interpreted mainly as associations rather than direct causal effects.

Assumptions

The research assumes that facial expressions provide a useful indication of emotional states and that the selected five emotion categories can be meaningfully recognized across different tourist groups.

It is also assumed that detected emotional states have a meaningful relationship with visitor satisfaction. The study further assumes that tourists will provide informed consent and voluntarily use the application when the purpose of data collection and privacy protections are clearly explained.

From a technical perspective, the research assumes that most target tourists have smartphones capable of running the application and that sufficient GPS and network connectivity will normally be available. When connectivity is limited, emotion recognition is expected to continue on the device while cloud-based processing and dashboard updates may be delayed.

The research also assumes that destination managers can define meaningful geographical zones and have the authority and resources to respond to system recommendations.

Finally, the study assumes that the Adaptive Emotion Recognition component can be evaluated as an individual research component even though it may operate alongside other tourism system functions. Results from the single-site pilot are considered initial evidence and may require validation across multiple destinations and longer periods in future research.

1.8 Individual contribution

This thesis presents the individual research contribution carried out as part of the broader group project R26-IT-112. The main contributions of this research are presented below.

Research Contributions

1. Mobile-Based Emotion Recognition for Tourism

This research develops a smartphone-based real-time emotion recognition approach for tourism destination management. Unlike previous approaches that depend on fixed cameras or wearable sensors, the proposed approach uses tourists' smartphones to capture and process facial expressions. This reduces infrastructure requirements and supports deployment in outdoor tourism locations.

2. Contextual Multi-Modal Emotion Recognition

The research combines facial emotion recognition with contextual information such as GPS zones, crowd density, and time. This approach improves the ability of the system to operate in outdoor environments without requiring additional specialized sensors.

3. Emotion-Based Recommendation System

The research extends emotion recognition from detection to action by providing recommendations for both tourists and destination managers. The system can suggest itinerary changes for tourists, provide zone-specific recommendations to managers, and priorities detected anomalies.

4. Validation of Multi-Modal Fusion

The experimental results show that contextual multi-modal fusion improved emotion classification accuracy from 87.3% for the facial-only approach to 91.7%. This provides empirical evidence of the benefit of combining facial and contextual information in a tourism environment.

5. Privacy-Preserving Architecture

The system follows a privacy-focused approach by processing facial information locally on the user's smartphone. Raw facial images are not transferred to the cloud; only anonymized emotion-related features are transmitted for further processing.

Technological Contributions

The research resulted in a functional mobile application using React Native, TensorFlow Lite, GPS-based zone mapping, and backend services. A practical late-fusion pipeline was implemented to combine different data sources while maintaining suitability for mobile devices.

In addition, a web-based dashboard was developed for destination managers. It provides emotion heatmaps, anomaly alerts, and recommendation information to support destination management decisions.

Practical and Industry Contributions

A pilot deployment at a Sri Lankan heritage site demonstrated the practical use of the system with tourists over a four-week period. The results also showed practical value for destination managers, with 80% considering anomaly alerts valuable and 60% finding the recommendations actionable.

The collected anonymized emotional data, including emotional states, locations, temporal patterns, and satisfaction relationships, can support future research. The work also provides a potential pathway for developing the system into a SaaS platform for tourism destination operators.

Contribution to Broader Knowledge

This research demonstrates how emotion recognition, contextual data, and tourism management can be combined in a practical system. It also provides a foundation for future research on tourism-specific emotions, demographic differences, long-term visitor behavior, and applications across different types of tourism sites. The

approach can further support more responsive and visitor-centered destination management.

1.9 Thesis structure

This thesis is organized into nine main sections.

Section 1: Introduction

Introduces the research background, problem statement, objectives, and overall research context.

Section 2: Literature Review and Related Work

Reviews existing research on emotion recognition, facial expression analysis, real-time emotion detection, multi-modal fusion, and tourism applications. The review is used to identify the research gaps addressed by this study.

Section 3: Research Methodology

Describes the Design Science Research approach, datasets, experimental procedures, evaluation methods, ethical considerations, privacy measures, and research documentation.

Section 4: Product and Solution Design

Presents the system architecture, requirements, design decisions, and main modules, including emotion recognition, multi-modal fusion, trajectory analysis, anomaly detection, and the manager dashboard.

Section 5: Implementation, Integration and Testing

Describes the technologies and development environment used to implement the system, together with integration, testing, and deployment activities.

Section 6: Results and Validation

Presents the main research and technical results, including the approximately 82–85% emotion recognition accuracy, multi-modal fusion results, system performance, and feedback from tourists and destination managers.

Section 7: Discussion

Discusses the research findings in relation to previous studies, research contributions, technical outcomes, limitations, and implications for tourism practice and future research.

Section 8: Commercialization and Sustainability

Discusses the value proposition, target users, market opportunity, business model, sustainability aspects, intellectual property, regulatory considerations, and relevant Sustainable Development Goals.

Section 9: Conclusion and Future Directions

Summarizes the research, evaluates the achievement of the objectives, presents the main contributions, and identifies possible directions for future development.

The thesis concludes with the references and appendices containing supporting materials such as ethical documentation, detailed test results, and technical specifications. This structure combines the theoretical aspects of the research with its practical implementation and real-world validation.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

A systematic literature review was conducted to identify existing research related to emotion recognition in tourism. The main databases used were Google Scholar, IEEE Xplore, and Scopus. The search used keywords such as *facial emotion recognition, facial expression analysis, emotion detection, affective computing,* and *emotion classification*, combined with tourism-related terms such as *tourism management, heritage sites, visitor experience,* and *destination management*. Searches also included terms related to LSTM, temporal modelling, GIS, zone mapping, and location-based analysis.

The review mainly considered peer-reviewed research published between 2014 and 2024. Earlier foundational studies were also considered where they were important to the theoretical background. Papers were selected when they focused on emotion recognition, facial expression analysis, affective computing, or tourism-related applications. Studies without sufficient experimental validation, duplicate publications, and research unrelated to the main research area were excluded.

The initial search produced more than 1,200 results. After title and abstract screening, around 120 papers were selected for detailed review, and ten core studies were identified as directly relevant to this research. The selected studies were examined based on their methodology, evaluation methods, findings, limitations, and future research directions.

The review covers five main areas: emotion recognition theory, deep learning for emotion recognition, multimodal emotion recognition, real-time and real-world emotion recognition, and tourism and hospitality applications.

2.2 Domain and theoretical foundations

Emotion Theory and Tourism

The theoretical background of emotion recognition is mainly based on Ekman's universal emotion model. His research showed that several basic emotions can be recognized through facial expressions across different cultures. These include happiness, sadness, fear, anger, disgust, and surprise.

For this research, five categories are considered: **anger, happiness, neutral, sadness, and surprise**. These categories provide a practical representation of positive, negative, and neutral visitor emotional states. In tourism, emotions are important because they can influence visitor satisfaction, future visits, and

recommendations. Tuerlan et al. [1] highlighted the strong relationship between customer emotions, satisfaction, and behavioral intentions in hospitality and tourism.

Deep Learning and CNN

Convolutional Neural Networks (CNNs) are widely used for facial expression and emotion recognition because they can automatically learn discriminative visual features from facial images. Convolutional layers learn low-level features such as edges, textures, and facial contours, while deeper layers learn higher-level representations associated with different facial expressions.

For the proposed system, a custom Deep Convolutional Neural Network was developed for facial emotion classification using the RAF-DB dataset. The architecture consists of multiple Conv2D layers for feature extraction, MaxPool2D layers for spatial dimensionality reduction, dropout layers for regularization, and dense fully connected layers for classification. The final classification stage produces emotion probabilities for the five target classes: anger, happiness, neutral, sadness, and surprise.

The custom CNN was selected as the facial emotion classification model because its architecture can be specifically configured for the target RAF-DB emotion classification task. The trained model was exported to TensorFlow Lite format to support efficient inference within the application environment.

Multimodal Emotion Recognition

Using more than one source of information can improve emotion recognition. Multimodal systems may combine facial expressions, voice, movement, and contextual information. Fusion can be performed using early fusion, late fusion, or hybrid fusion.

Late fusion processes each source separately and combines the final predictions. This approach provides flexibility when one source is unavailable. Kalateh et al. [3] identified late fusion as a practical approach for multimodal systems, while Khare et al. [4] reported that multimodal approaches can improve recognition compared with single-modality systems.

Contextual information such as location, crowd level, and time can also support emotion recognition, especially in outdoor environments. Ezzameli and Mahersia [5] highlighted the performance difference between laboratory and real-world conditions and the importance of contextual information.

Temporal Emotion Analysis

Emotions can change throughout a visitor's experience. Therefore, analyzing individual emotion predictions alone may not provide the complete picture. Long Short-Term Memory (LSTM) networks are suitable for analyzing sequential data

because they can learn relationships between previous and current emotional states. LSTM-based temporal modelling can help identify patterns such as continuous engagement, emotional decline, or recovery [3], [6].

Attention mechanisms can also identify important moments in an emotional sequence. This can help determine when a visitor's emotional state changes significantly.

Anomaly Detection

Anomaly detection can identify unusual emotional patterns. For example, a sudden increase in negative emotions or continuous negative emotions in a particular zone may indicate a problem.

Rule-based anomaly detection is useful for tourism applications because the rules can be understood and adjusted by destination managers. Examples include detecting sustained negative emotions, sudden emotional changes, or unusually high negative emotions in a specific zone.

2.3 Related research

Several studies have investigated emotion recognition in tourism and related fields.

González-Rodríguez et al. [7] investigated facial expression recognition as a method for measuring tourist satisfaction. Their study showed a relationship between facial emotions and self-reported satisfaction. However, the research was mainly laboratory-based and did not provide mobile deployment, spatial mapping, temporal tracking, or recommendations. This shows the gap between laboratory-based emotion recognition and real tourism environments.

Kwon and Yu [8] developed a real-time multi-CNN system for evaluating museum visitor satisfaction. The system used fixed cameras and achieved real-time processing with reported accuracy around 88–91%. It demonstrated that emotion recognition can work in an actual museum environment. However, it depended on fixed cameras and did not provide GPS-based zone mapping, emotional trajectory analysis, or a recommendation system.

Matsuda et al. [9] developed EmoTour, a multimodal system that used wearable sensors, audiovisual information, and behavioral data to estimate tourist emotions and satisfaction. The study showed that combining multiple sources improved recognition. However, the use of wearable devices creates problems related to scalability, maintenance, charging, and deployment for large numbers of tourists. The system also did not provide spatial mapping or a closed-loop recommendation mechanism.

Qi and Han [10] investigated emotion recognition for emergency situations in tourism using an improved CNN. Their system focused on detecting fear and extreme distress to support emergency response. Although this demonstrates the value of emotion recognition for safety, it mainly provides reactive emergency detection rather than continuous preventive monitoring. It also does not provide zone mapping, emotional trajectories, or proactive recommendations.

Khare et al. [4] conducted a systematic review of emotion recognition research from 2014 to 2023. Their findings showed that multimodal systems generally perform better than unimodal systems, while outdoor and uncontrolled environments remain challenging. The review also highlighted demographic variation, real-time processing, and mobile deployment as important challenges. Tourism applications were identified as an area that requires further research.

Kalateh et al. [3] reviewed multimodal emotion recognition methods and found that late fusion is practical for real-world systems because each modality can be processed independently. The study also highlighted LSTM and transformer-based approaches for temporal analysis. However, contextual information and mobile deployment remain less explored, particularly in tourism.

Ezzameli and Mahersia [5] reviewed the development of emotion recognition from unimodal to multimodal systems. They identified a significant performance gap between laboratory and real-world environments, especially for facial-expression-only systems. Outdoor conditions, demographic differences, and environmental context can affect recognition performance. Their findings support the need for multimodal and context-aware systems.

Tuerlan et al. [1] reviewed customer emotion research in hospitality and tourism and identified a strong relationship between emotions, satisfaction, and loyalty. They also highlighted the limitations of traditional post-visit surveys because they cannot provide real-time information during the visitor experience. This supports the need for technology-based, continuous emotion measurement.

Kement et al. [11] analyzed facial emotion recognition research in hospitality and tourism. Their review found that most existing studies were theoretical or laboratory-based, with limited real-time deployment. They also identified a lack of systems that connect emotion recognition with actionable recommendations for visitors and destination managers.

Santamaria-Granados et al. [12] developed a tourist recommendation system using emotion recognition from wearable physiological data. The study showed that emotional information can improve personalized recommendations. However, wearable devices create scalability and maintenance problems, and the system did not provide zone-level emotion mapping or manager-focused operational recommendations.

Overall, these studies demonstrate that emotion recognition can support tourism and visitor experience analysis. However, most existing systems focus on individual aspects such as facial recognition, multimodal sensing, emergency detection, or recommendations rather than combining them into one deployable tourism system.

2.4 Existing systems products and techniques

Several commercial and open-source technologies are available for emotion and facial analysis. Commercial systems such as Affectiva, Microsoft Azure Emotion API, Amazon Rekognition, and Google Cloud Vision have provided facial or emotion-related analysis. However, many of these systems depend on cloud processing and are not specifically designed for tourism. They also do not directly provide tourism-based spatial mapping, temporal analysis, or recommendation functions.

Open-source technologies such as TensorFlow, Keras, MediaPipe, DeepFace, and OpenFace provide useful components for facial analysis. MediaPipe is particularly useful for real-time facial landmark detection and can support lightweight perception pipelines [13]. However, these tools generally provide individual technical functions rather than a complete tourism emotion management system.

Different techniques can be used to improve the proposed system. CNN-based models can perform facial emotion classification, while Custom CNN provides a lightweight option for mobile deployment [2]. Data augmentation can improve robustness against lighting changes, head movements, and partial occlusion. Model optimization methods such as quantization, pruning, and knowledge distillation can further reduce computational requirements.

For temporal analysis, LSTM networks can model emotion sequences, while attention mechanisms can help identify important moments in a visitor's emotional trajectory [6]. For multimodal integration, weighted averaging, learned fusion layers, and ensemble methods can be considered.

2.5 Gap synthesis and conceptual framework

The literature review shows that the main research gap is not the absence of individual technologies. Facial emotion recognition, mobile deep learning, temporal modelling, and recommendation systems already exist separately. The main gap is the lack of their integration into a practical and deployable system for tourism destination management.

Gap 1: Mobile Platform Deployment

Existing real-time systems mainly depend on fixed cameras or wearable devices [8], [9], [12]. Fixed cameras require infrastructure, while wearable devices are difficult to distribute and maintain for large numbers of tourists.

This research addresses this gap through a **smartphone-based emotion recognition approach**, reducing the need for additional destination infrastructure.

Gap 2: Zone-Level Spatial Emotion Mapping

Existing studies can measure visitor emotions but generally do not connect them to specific physical locations. For example, museum-based systems can detect visitor emotions but do not identify the exact zone or exhibit related to the emotional response [8].

This research addresses the gap by using **GPS-based zone identification** so that emotional information can be analyzed according to specific destination areas.

Gap 3: Temporal Emotional Trajectory Tracking

Most existing systems provide individual or snapshot emotion classifications rather than analyzing how emotions change throughout a visit. Reviews have highlighted the importance of temporal information and sequence modelling [4], [3].

Gap 4: Anomaly Detection for Preventive Intervention

Qi and Han [10] focused on detecting emotional states related to tourism emergencies. However, this represents mainly a reactive approach. There is a need to identify negative emotional patterns before they become serious problems.

This research addresses the gap through **rule-based anomaly detection**, including sustained negative emotions, sudden emotional changes, and unusual negative patterns within specific zones.

Gap 5: Closed-Loop Recommendation Engine

Existing research shows that emotion information can support tourism recommendations [12]. However, most emotion recognition systems stop after detecting and analyzing emotions. Kement et al. [11] also identified the lack of systems that connect emotion information with actionable tourism recommendations.

This research addresses the gap by providing a **closed-loop recommendation mechanism** for both tourists and destination managers. It can provide personalized suggestions for tourists and identify zone-specific issues that require management attention.

Conceptual Framework

The proposed conceptual framework combines the main findings from the literature into three layers.

Theoretical Foundation:
Emotion theory provides the basis for facial expression analysis, while tourism research establishes the relationship between emotions, satisfaction, and visitor behavior [1]. Research on multimodal emotion recognition supports the use of multiple information sources for better real-world performance [5], [4].

Technical Architecture:
The custom CNN model provides the lightweight deep learning model for mobile emotion recognition [2]. Late fusion is used to combine facial emotion with contextual information such as GPS zone and crowd conditions [3]. LSTM is used for temporal emotion analysis [3], [6], while rule-based anomaly detection identifies important negative patterns.

Application Layer:
The system connects emotion recognition with tourism management and visitor experience. It provides real-time information for visitors and supports destination managers in identifying areas that may require attention. The approach also supports data-driven and sustainable tourism management.

The main integration points are:

1. **Mobile deployment + zone mapping** through smartphone and GPS information.
2. **Emotion recognition + temporal analysis** through LSTM-based sequence processing.
3. **Temporal patterns + anomaly detection** through rule-based analysis.
4. **Emotion analysis + decision-making** through the recommendation mechanism.

Overall, the framework combines affective computing, tourism management, and mobile system technologies into one system. It addresses the five identified gaps by integrating emotion recognition, spatial information, temporal analysis, anomaly detection, and recommendations into a single tourism-focused solution.

3 RESEARCH METHODOLOGY

3.1 Research design

Design Science Research Methodology

This research follows the **Design Science Research (DSR)** methodology because the main objective is to design, develop, and evaluate an Adaptive Emotion Recognition system for tourism environments. DSR is suitable for this research as it focuses on developing a technological artifact to address an identified real-world problem.

The research follows five main DSR phases.

Phase 1 - Problem Identification and Motivation:

The research problem was identified through the literature review, analysis of existing tourism emotion recognition systems, and identification of five research gaps presented in Section 2.5. These gaps provided the main motivation for developing the proposed system.

Phase 2 - Objective Definition:

The identified gaps were converted into research objectives and measurable evaluation criteria. These included emotion classification accuracy, processing latency, multimodal fusion performance, anomaly detection, usability, and system feasibility.

Phase 3 - Design and Development:

The Adaptive Emotion Recognition component was designed and implemented as part of the proposed system. The main technologies included React Native for the mobile application, Custom CNN with TensorFlow Lite for on-device emotion recognition, GPS-based zone identification, multimodal data processing, anomaly detection, and a web-based manager dashboard.

Phase 4 - Demonstration and Evaluation:

The developed component was evaluated using the RAF-DB dataset, representative mobile devices, and a pilot deployment. Technical performance, usability, system feasibility, and user feedback were considered during evaluation.

Phase 5 - Communication:

The research findings, system design, implementation, evaluation results, and limitations are documented in this thesis.

Mixed-Methods Approach

Although DSR is the main methodology, both quantitative and qualitative methods were used.

Quantitative evaluation included emotion classification accuracy, precision, recall, F1-score, processing latency, battery consumption, anomaly detection performance, and usability scores.

Qualitative data were collected through tourist feedback and interviews with destination managers. These data helped identify user perceptions, system limitations, recommendation usefulness, and practical deployment issues.

Therefore, the mixed-methods approach provided both technical performance measurements and practical feedback about the proposed system.

3.2 Data participants or study materials

Training and Test Dataset

The main dataset used to train and evaluate the emotion recognition model was the **RAF-DB (Real-world Affective Faces Database)**. It contains facial images collected from real-world Internet sources and therefore provides more variation than controlled laboratory datasets.

RAF-DB contains different facial expressions under variations in head pose, lighting, image quality, occlusion, and expression intensity. These characteristics are useful for the proposed tourism application because tourists may be captured under different outdoor and indoor conditions.

The dataset was divided into training, validation, and test sets. The training set contained 12,271 images, the validation set contained 3,068 images, and the test set contained 3,268 images.

The original proposal considered FER2013. However, RAF-DB was selected because its real-world images and variation were considered more suitable for the intended deployment environment.

The model used five emotion classes: **anger, happiness, neutral, sadness, and surprise**. The dataset contained an imbalance between classes, with happiness and neutral having more samples than some other emotions. Therefore, weighted F1-score was also used together with accuracy during evaluation.

For CNN-based emotion classification, the detected facial region is cropped and resized to 100 × 100 pixels. The cropped image is converted to grayscale and the

pixel values are normalized to the [0, 1] range before being provided to the CNN model.

Pilot Participants

The pilot study involved **5 tourists** visiting a selected Sri Lankan heritage site. Participants were adults who were able to use a smartphone and provide informed consent. Participation was voluntary, and participants could stop using the application at any time.

A further **1 destination managers** participated in evaluating the manager dashboard and system recommendations. They were selected because of their involvement in visitor experience and operational management at the pilot site.

Study Materials

The main study materials consisted of:

- React Native mobile application
- Custom CNN-based emotion recognition model/TensorFlow Lite emotion recognition model
- GPS-based zone mapping
- Backend REST API
- PostgreSQL database
- Rule-based anomaly detection module
- Destination manager dashboard
- Tourist exit survey
- Manager interview and feedback forms

The mobile application collected emotion class, confidence, zone, and timestamp information. Raw facial images were processed on the device and were not stored or transmitted to the backend.

3.3 Experimental or study procedure

Overall System Procedure

The individual Adaptive Emotion Recognition component follows a sequence from data capture to analysis and recommendation.

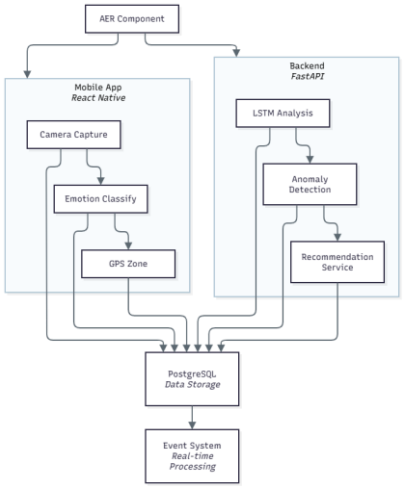


Figure 1: Adaptive Emotion Recognition System - Individual Component

This figure represents only the **individual Adaptive Emotion Recognition component**, rather than the complete research system.

Model Training Procedure

The facial emotion recognition model was developed using a custom Deep Convolutional Neural Network (CNN) trained on the RAF-DB dataset. The architecture consists of multiple Conv2D layers for extracting facial features, MaxPool2D layers for reducing spatial dimensions, dropout layers for regularization, and dense fully connected layers for final classification.

Before CNN inference, facial regions are detected using an OpenCV Haar Cascade classifier. The largest detected face is selected and cropped from the input image. To improve robustness to tilted faces, the detection pipeline also attempts rotations of −15° and +15° when a suitable face is not detected in the original image.

The detected facial crop is resized to 100 × 100 pixels, converted to grayscale, and normalized to the [0, 1] range. These preprocessing steps produce the input representation expected by the custom CNN.

To improve prediction robustness, a crop-ensemble inference strategy is used. Three forward passes are performed using crop scaling factors of 0.95, 1.0, and 1.05. The resulting probability vectors are averaged arithmetically to obtain the final emotion probability distribution and predicted emotion class.

The model was trained using the RAF-DB dataset and subsequently exported to TensorFlow Lite format to provide a lightweight model representation for inference.

The main training settings were:

Parameter	**Value**
Model	Custom Deep CNN
Dataset	RAF-DB
Emotion Classes	5
Architecture	Conv2D + MaxPool2D + Dropout + Dense
Face Detection	OpenCV Haar Cascade
Maximum epochs	50
Model Format	TensorFlow Lite (.tflite)
Loss function	Categorical cross-entropy
Validation Accuracy	Approximately 82–85%

Table 1: Model Training Parameters

The model was trained using the training set and validated after each epoch. Early stopping was used to reduce overfitting. The final model was then evaluated using the independent test set.

The trained model achieved approximately **82-85% validation accuracy test accuracy** before mobile optimization.

Mobile Model Optimization

After training, the model was converted into TensorFlow Lite format for mobile deployment. Post-training 8-bit quantization reduced the model size from approximately 13 MB to 3.2 MB. The optimized model achieved approximately **90.9% accuracy**, while providing a smaller model suitable for mobile devices.

Mobile Application Development

The mobile application was developed using React Native. The application included camera access, emotion detection, GPS-based zone identification, local data storage,

backend communication, recommendation display, and a post-visit emotional summary.

Development was carried out in stages:

1. Mobile application setup and permission handling
2. Camera and real-time emotion recognition
3. GPS-based zone mapping
4. Local data aggregation and backend communication
5. User interface implementation
6. Testing and performance optimization

The application processed emotion recognition on the device. Emotion summaries were transmitted to the backend at regular intervals rather than continuously sending camera frames.

The application was tested on representative iOS and Android devices. Measured processing latency was approximately **78 - 145 ms**, depending on the device, and all tested devices remained within the target of 150 ms per frame.

Pilot Deployment

A Sri Lankan heritage site with several visitor zones was selected for the pilot deployment. The site was divided into predefined geographical zones, such as the entrance area, main temple complex, meditation garden, exhibition area, and visitor facility area.

The pilot was conducted over four weeks.

- **Week 1:** Baseline visitor satisfaction data were collected without the proposed system.
- **Weeks 2-3:** The mobile application was used by participating tourists and the emotion recognition and dashboard functions were active.
- **Week 4:** Tourist surveys and manager interviews were conducted and system performance was reviewed.

During the pilot, emotion information was processed on the mobile device and summarized data were transmitted to the backend. Managers could view zone-level emotional information, anomaly alerts, and recommendations through the dashboard.

Backend Processing

The backend was implemented using a REST API and PostgreSQL database. The system stored anonymized emotion records containing emotion class, confidence, zone, timestamp, and session information.

The LSTM module analyzed emotional sequences to identify common trajectory patterns, while rule-based anomaly detection identified conditions such as sustained negative emotions, sudden emotional changes, and high negative emotion levels within a zone.

The dashboard provided managers with zone-level heatmaps, time-series information, anomaly alerts, and recommendations.

3.4 Measures and evaluation metrics

The developed component was evaluated using technical, usability, and practical performance measures.

Emotion Classification

Overall accuracy was calculated using the held-out RAF-DB test set. Precision, recall, and F1-score were also calculated for each emotion class because the dataset contained class imbalance.

The main target for the emotion classification model was **90% or higher accuracy**.

Processing Latency

Processing latency was measured from camera frame capture to the availability of the emotion classification result.

Mean latency, 95th percentile latency, maximum latency, and standard deviation were considered. The target was **less than 150 ms per frame** to support near real-time operation.

Multimodal Fusion

The facial-only model was compared with the multimodal approach using contextual information such as zone and temporal data.

The comparison examined whether the addition of contextual information improved emotion-related analysis and reduced inappropriate predictions.

Anomaly Detection

Anomaly detection was evaluated using precision and recall. The target values were approximately **85% precision** and **80% recall**.

Three main anomaly conditions were considered:

1. Sustained negative emotions within a zone
2. Sudden changes in emotional state
3. High levels of negative emotions within a zone

Usability

The **System Usability Scale (SUS)** was used to evaluate the mobile application and manager dashboard. The target was a score of at least 70 out of 100.

Tourists were also asked about perceived emotion detection accuracy, while managers provided feedback on the clarity and actionability of system recommendations.

System Performance

Battery consumption was measured during continuous two-hour use on representative devices. Backend scalability was also tested by increasing the number of concurrent users and monitoring response latency and resource usage.

Pilot Impact

The pilot compared visitor satisfaction during the baseline and active deployment periods. The analysis considered satisfaction ratings, staff interventions, anomaly alerts, and implemented recommendations

3.5 Data analysis

Quantitative Analysis

Quantitative data were analyzed using descriptive statistics and performance measures.

For emotion recognition, a confusion matrix was used to identify correctly and incorrectly classified emotions. Precision, recall, and F1-score were calculated for each emotion class.

The main misclassification patterns were examined to identify confusion between similar expressions, particularly between happiness and neutral expressions.

For the pilot study, visitor satisfaction results from the baseline and deployment periods were compared using descriptive statistics and a two-sample t-test. Statistical significance was considered at $p < 0.05$, together with effect size where appropriate.

Qualitative Analysis

Manager interviews and open-ended tourist feedback were analyzed using thematic analysis.

The main themes included:

- Actionability of recommendations

- Trust in the system
- Operational impact
- System limitations
- Possible future improvements

This analysis helped identify practical issues that could not be fully represented through numerical performance measures.

Error Analysis

Incorrect emotion predictions were reviewed to identify common causes such as ambiguous expressions, lighting conditions, facial occlusion, head pose, and possible demographic differences.

The findings were used to identify possible future improvements, including additional training data, stronger augmentation, confidence thresholding, and model improvement.

3.6 Ethics privacy security and responsible AI

Ethical approval was obtained before conducting the pilot study, and participant involvement was based on informed consent.

Participation was voluntary, and participants were informed about the purpose of the research, the data collected, how the data would be processed, and their right to withdraw.

Privacy Protection

A major privacy feature of the system is **on-device emotion processing**. Facial images are processed locally using the TensorFlow Lite model and are not transmitted to the backend.

Only summarized information such as emotion class, confidence, zone, timestamp, and anonymized session ID is transmitted.

Raw facial images are deleted from memory after processing and are not stored on the server.

Data Protection

The collected data were protected using encryption during transmission and storage. TLS 1.3 was used for communication between the mobile application and backend services, while stored data were protected using encryption and access controls.

Role-based access control was used to restrict access. Researchers could access anonymized research data, while destination managers could only access aggregated information relevant to their site.

Data were planned to be retained for a maximum of 12 months and then deleted.

Responsible AI

The system was designed to provide emotion estimates rather than claiming to identify a person's actual emotional state with certainty. Confidence scores were displayed with predictions, and users were informed that the system could make errors.

The system did not automatically make operational decisions. Instead, recommendations and anomaly alerts were provided to managers, who retained responsibility for deciding whether an intervention was appropriate.

Potential demographic bias was also considered by examining performance across available demographic groups. Differences in model performance were treated as an area for further investigation and improvement.

Consent

Participants received information about camera and GPS usage, emotion processing, data storage, privacy protection, and withdrawal procedures before using the application.

Participants could stop using the application at any time. Requests for deletion of collected data could also be made according to the study procedure.

3.7 Reproducibility and research artefacts

To support reproducibility, the research documents the main model architecture, training parameters, preprocessing methods, application structure, and evaluation procedures.

The main research artifacts include:

- Trained Custom CNN emotion recognition model
- TensorFlow Lite mobile model
- Model training scripts
- React Native application source code
- Backend API implementation
- Database structure

- Zone definitions
- Evaluation results
- Survey and interview materials
- System and deployment documentation

The training process can be reproduced using the RAF-DB dataset and the documented preprocessing and training parameters.

The mobile application and backend structure are also documented so that the individual component can be rebuilt and tested on supported devices.

However, exact reproduction of the pilot results is not always possible because real-world factors such as tourist behavior, location, weather, time period, device hardware, and site conditions can change.

Therefore, future researchers can reproduce the technical experiment using the same dataset and configuration while conducting new pilot studies at different tourism locations to evaluate generalizability.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

The requirements define the main functions and quality expectations of the Adaptive Emotion Recognition (AER) system. They are divided into functional and non-functional requirements.

4.1.1 Functional Requirements

FR1: Real-Time Facial Emotion Classification

The system shall capture facial expressions through the smartphone camera and classify them into five emotion categories: **anger, happy, neutral, sad, and surprise**.

The mobile application captures frames at 10 FPS. Each frame is resized to 100 * 100 pixels, normalized, and processed using the Custom CNN emotion recognition model. The system returns the predicted emotion and confidence score. The classification latency should be below 150 ms per frame.

If a face cannot be detected, the system continues attempting classification. After three consecutive failures within 10 seconds, a message is shown asking the user to position the camera clearly.

Success criteria:

- Minimum 90% classification accuracy on the test set.
- Mean classification latency below 150 ms.
- Operation under different lighting, head poses, and partial occlusion conditions.

FR2: GPS-Based Zone Identification

The system shall identify the tourist's current location using GPS and assign the tourist to predefined heritage-site zones.

GPS coordinates are updated approximately every 5 seconds and matched with predefined GeoJSON zone polygons. Point-in-polygon processing is used for zone assignment. If GPS is temporarily unavailable, the last known zone is retained for up to two minutes before the location is marked as unknown.

Each emotion record is linked with the zone ID and timestamp.

Success criteria:

- Zone assignment accuracy of at least 95%.
- Zone assignment within approximately 5 seconds of receiving a GPS update.

- Graceful operation when GPS is unavailable.

FR3: Multi-Modal Data Fusion

The system shall combine facial emotion information with contextual information to improve emotion interpretation. The main modalities are:

1. Facial emotion probabilities from Custom CNN.
2. Current GPS zone and historical zone emotion distribution.
3. Crowd density estimated using person detection.
4. Time of day and duration spent in the current zone.

A late-fusion approach is used to combine these inputs. Facial emotion receives the highest weight because it is the direct emotion signal, while zone, crowd, and temporal information act as contextual factors.

The initial design uses weights of approximately **0.6, 0.2, 0.1, and 0.1** for facial, zone, crowd, and temporal information respectively. The weights can be optimized during training.

Success criteria:

- Fused model accuracy at least 5% higher than the facial-only baseline.
- Fusion processing below 30 ms.
- Continued operation when one contextual modality is unavailable.

FR4: Temporal Emotional Trajectory Analysis

The system shall analyses changes in tourist emotions throughout a visit. Emotion data is aggregated into one-minute records and used to identify longer-term emotional patterns.

Trajectory analysis is used to analyses the sequence. The system identifies five trajectory patterns:

- Sustained Positive Engagement
- Gradual Decline
- Fluctuating Engagement
- Sustained Negative
- Recovery Pattern

Trajectory analysis is performed during the visit, at the end of the session, and through overnight analysis.

Success criteria:

- At least 80% trajectory classification accuracy.
- Identified patterns should show a relationship with post-visit satisfaction.
- Critical emotional changes should be identifiable through the attention mechanism.
- End-of-session analysis should complete within 5 seconds.

FR5: Real-Time Anomaly Detection

The system shall identify unusual emotional patterns and generate alerts for possible intervention.

Three main anomaly conditions are used:

1. **Sustained Negative Emotion:** sad or angry emotions continue for more than three minutes in one zone.
2. **Sudden Emotion Shift:** emotional valence changes by two or more levels within 30 seconds.
3. **Zone Dissatisfaction:** more than 40% of visitors in a zone show negative emotions within a 30-minute period.

To reduce false alerts, anomaly detection considers confidence scores above 0.70 and repeated observations rather than a single frame. Duplicate alerts for the same zone and anomaly type are suppressed for 15 minutes.

Alerts can be delivered through the tourist application, manager dashboard, and optionally SMS.

Success criteria:

- Precision above 85%.
- Recall above 80%.
- Alerts delivered within 30 seconds.
- False-positive rate below 10% based on manager evaluation.

FR6: Personalized Tourist Recommendations

The system shall generate recommendations when a tourist experiences a detected anomaly or remains neutral/negative for an extended period.

Recommendations consider:

- Current emotion.
- Current zone.

- Time of day.
- Visit duration.
- Previously visited zones.
- Historical satisfaction of other zones.

For example, if a tourist shows negative emotions in one zone, the system can recommend a nearby zone that has historically produced more positive emotional responses. The tourist can accept, dismiss, or provide feedback on the recommendation.

Success criteria:

- At least 40% of recommendations acted upon.
- At least 60% rated as helpful.
- Recommendations adapt to current conditions rather than remaining static.

FR7: Destination Manager Dashboard

The system shall provide a web-based dashboard for destination managers. The dashboard presents:

- Real-time zone emotion heatmaps.
- Time-series satisfaction trends.
- Anomaly logs.
- AI-generated recommendations.
- Zone and time-based comparisons.
- Historical analytics.

The dashboard uses role-based access, authentication, MFA, and automatic session timeout. Managers can export reports and configure alert types.

Success criteria:

- Manager SUS score of at least 70/100.
- Recommendations rated as actionable by at least 60% of managers.
- Dashboard loading time below 3 seconds.
- Emotion information updated within 60 seconds.

4.1.2 Non-Functional Requirements

NFR1: Performance and Real-Time Processing

The system requires mean emotion classification latency below **150 ms**, with 95% of frames processed within 200 ms.

Testing on selected devices showed the following approximate results:

The backend API should respond within 200 ms for normal dashboard operations. The system is designed to support at least **500 concurrent users**, with cloud autoscaling based on system load.

NFR2: Battery Consumption

The mobile application should consume less than **15% battery per hour** during continuous operation. To reduce battery usage, the system uses adaptive processing, local data aggregation, batch transmission, and mobile hardware acceleration where available.

Measured results included approximately 14% per hour on an iPhone 12, 15.5% on a Pixel 5, and 19% on an iPhone 8.

NFR3: Data Security

Emotion data is treated as sensitive information. Communication uses **TLS 1.3**, while stored data and backups use encryption. AWS KMS was proposed for encryption key management in a future AWS deployment.

Role-based access control, MFA, JWT/API authentication, and session timeout are used to protect the system. Raw facial images are processed on the device and are not stored on the backend.

NFR4: Reliability and Availability

The target system availability is **99% or higher**. Critical backend services use redundancy, database replication, load balancing, and automated backups.

The mobile application can continue emotion classification when the network is unavailable because the TensorFlow Lite model operates locally. Data can be stored temporarily and transmitted when connectivity returns.

NFR5: Scalability

The application is designed to operate on devices with at least 2 GB RAM. The mobile model is targeted at a size below 5 MB, and application startup should take less than 2 seconds.

The backend is designed for at least 500 concurrent users. Database indexing, caching, autoscaling, and data-retention policies are used to support increasing data volumes.

NFR6: Usability

The mobile application and manager dashboard target a **SUS score of at least 70/100**.

The application should be simple enough for first-time users to understand within approximately five minutes. Accessibility features such as larger text, suitable contrast, screen-reader support, and responsive interfaces are considered.

NFR7: Maintainability and Deploy ability

The system follows consistent coding standards and uses Git for version control. Unit and integration testing are applied to important components, with a target of at least 80% code coverage.

Terraform was considered for the proposed AWS deployment but was not used in the current prototype, while CI/CD supports automated testing and deployment. Application logs and performance information are monitored through cloud monitoring tools.

4.2 Component architecture within the complete solution

The AER component is part of the broader **Next-Generation AI Framework for Intelligent Travel Support in Sri Lanka (R26-IT-112)**.

The complete framework contains the following major components:

- **Adaptive Emotion Recognition (AER):** Captures and analyses tourist emotions.
- **Personalized Journey Recommender (PJR):** Uses visitor behavior and emotion information to generate personalized journey suggestions.
- **Resource Optimization Component (ROC):** Uses crowd and satisfaction information to support staff and capacity management.
- **Shared Data Platform:** Provides common storage, APIs, event communication, and analytics.
- **Tourist Mobile Application:** Provides tourist-facing services.
- **Destination Manager Dashboard:** Provides operational information and alerts.

AER Component Architecture

The AER component contains the following major modules:

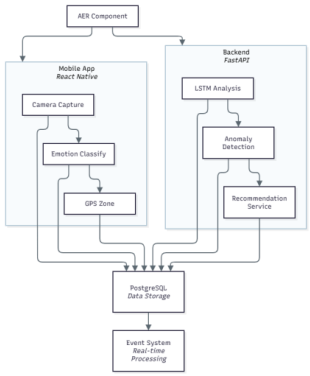


Figure 2:AER Component Architecture

The mobile layer performs camera capture, emotion classification, GPS processing, data aggregation, and user interaction. The backend manages data storage, trajectory analysis, anomaly detection, and recommendation generation.

The database stores emotion records, sessions, zones, trajectories, anomalies, and recommendations. A message queue can be used to publish emotion and anomaly events between services.

Data Flow

The main data flow is:

1. The tourist opens the mobile application and provides the required permissions and consent.
2. Camera frames are processed using the TensorFlow Lite custom CNN emotion recognition model.
3. GPS information is used to identify the current zone.
4. Emotion information is temporarily stored on the device and aggregated.
5. Aggregated information is securely sent to the backend.
6. The backend stores the information and performs anomaly and trajectory analysis.
7. When required, the recommendation service generates a suitable recommendation.
8. The manager dashboard receives updated emotion and anomaly information.
9. Emotion and anomaly information can also be shared with PJR and ROC through the common platform.

The AER component can operate independently because the main emotion classification is performed on the mobile device. However, integration with PJR and ROC provides additional functionality.

4.3 Design alternatives and decisions

Several design alternatives were considered before selecting the final architecture.

4.3.1 Emotion Classification Model

Three approaches were considered:

Alternative	Main Advantage	Main Limitation	Decision
ResNet-50	Higher possible accuracy	Large model and high latency	Rejected
Action Unit + Deep Learning	More interpretable	More complex processing	Rejected
Multiple CNN Ensemble	Improved robustness	Higher computation and battery use	Rejected
Custom CNN	Small, fast, suitable for Task-specific architecture and controllable model complexity	Lower generalization compared with some large pretrained models	**Selected**

A custom CNN was selected as the final facial emotion classification model. The model was specifically developed for the five-class RAF-DB emotion classification task and uses convolutional feature extraction, max-pooling, dropout regularization, and dense classification layers. The resulting model was exported to TensorFlow Lite format for lightweight inference. The preprocessing pipeline further improves robustness through face detection, rotation retries, grayscale normalization, and three-crop ensemble prediction.

4.3.2 Multi-Modal Fusion

Early fusion, hierarchical fusion, and late fusion were considered.

Early fusion requires all modalities to be available at the same time, while hierarchical fusion adds additional architectural complexity. **Late fusion was**

selected because individual modalities can operate independently and unavailable inputs can be handled without stopping the complete system.

The selected approach combines facial, zone, crowd, and temporal information using weighted outputs.

4.3.3 Anomaly Detection

Three approaches were considered:

- Machine-learning-based anomaly detection.
- Statistical threshold detection.
- Rule-based anomaly detection.

Machine-learning detection requires a large set of labelled anomaly examples, which was not available at the beginning of the study. Statistical thresholds can also be difficult to adapt to different tourism conditions.

Therefore, **rule-based anomaly detection was selected**. The rules are interpretable, configurable, and based on operational conditions such as sustained negative emotion, sudden emotional changes, and zone dissatisfaction.

4.3.4 Dashboard Design

Three dashboard approaches were considered:

- Continuous real-time streaming.
- Batch reporting.
- Near-real-time updates.

Continuous streaming increases system load and can make the dashboard difficult to interpret. Batch reporting provides useful historical information but is too slow for immediate intervention.

Therefore, a **near-real-time dashboard with approximately 60-second updates** was selected.

4.3.5 Database Selection

MongoDB, InfluxDB, and PostgreSQL with PostGIS were considered.

MongoDB provides flexible document storage, while InfluxDB is optimized for time-series information. However, the AER system requires both spatial and temporal queries.

PostgreSQL with PostGIS was selected because it supports relational data, spatial queries, indexing, transactions, and standard SQL. This makes it suitable for storing emotion records together with heritage-site zone information.

4.4 Detailed design

4.4.1 Mobile Application Design

The mobile application is developed using **React Native** and follows a simple MVC-based structure.

The main modules are:

- **Consent Module:** Handles user consent and privacy information.
- **Camera Module:** Captures facial images.
- **Emotion Classification Module:** Preprocesses images and runs the TensorFlow Lite model.
- **GPS and Zone Module:** Determines the current heritage-site zone.
- **Data Aggregation Module:** Stores and aggregates emotion records locally.
- **Recommendation Module:** Receives and displays recommendations.
- **Session Module:** Manages visit sessions and zone visits.

Emotion classification uses the custom CNN TensorFlow Lite model. Before inference, the facial region is detected using Haar Cascade, cropped, resized to 100 × 100 pixels, converted to grayscale, and normalized to the [0, 1] range. Three crop scales of 0.95, 1.0, and 1.05 are evaluated, and their probability vectors are averaged to obtain the final prediction.

GPS coordinates are compared with predefined GeoJSON zone boundaries using point-in-polygon processing. If GPS is temporarily unavailable, the most recent valid zone is retained for a limited period.

Raw facial images remain on the device and are not transmitted to the backend.

4.4.2 Backend Design

The backend is implemented using **FastAPI and Python**.

The main backend services are:

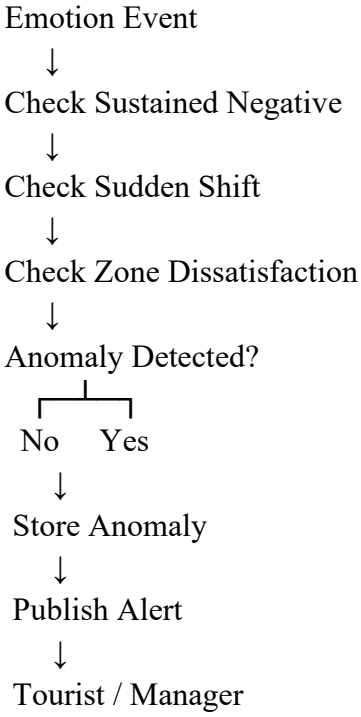
- Emotion data API.
- Zone satisfaction service.
- Recommendation service.
- Anomaly detection service.
- Dashboard API.

The main API operations include receiving emotion data, retrieving zone satisfaction, retrieving recommendations, recording recommendation feedback, and retrieving detected anomalies.

The backend stores information in PostgreSQL and communicates events between services through a message queue.

4.4.3 Anomaly Detection Design

The anomaly service monitors incoming emotion information and applies the predefined rules.



Confidence thresholds and repeated observations are used to reduce false alerts.

4.4.4 Manager Dashboard

The dashboard is developed using **React.js** and provides:

- Zone emotion heatmaps.
- Satisfaction trends.

- Anomaly history.
- Recommendation information.
- Comparative zone analysis.
- Historical reports.

The dashboard refreshes approximately every 60 seconds. Access is controlled through authentication, role-based permissions, and MFA.

The dashboard allows managers to identify areas with negative emotional patterns and review possible actions.

4.5 Standards and professional practice

The AER system follows common software engineering, security, testing, database, and accessibility practices.

4.5.1 Software Development Standards

The Python backend follows **PEP 8** coding standards. React Native and JavaScript code use consistent naming and project structure.

Git is used for version control with feature branches, meaningful commits, code reviews, and release versions.

Testing includes:

- Unit testing using pytest and Jest.
- Integration testing for APIs and databases.
- End-to-end testing for the mobile-to-backend workflow.

At least 80% code coverage is targeted for important components.

4.5.2 Security Standards

Security is considered throughout the system. The main practices include:

- TLS 1.3 for data transmission.
- Encryption for stored data.
- JWT/API authentication.
- Role-based access control.
- MFA for sensitive access.
- Secure session management.

- Parameterized database queries.
- Security and dependency monitoring.

The system also considers the **OWASP Top 10**, especially authentication, access control, injection, data protection, and logging.

4.5.3 API and Database Standards

The backend follows REST API principles with suitable HTTP methods and status codes. API documentation is maintained using OpenAPI/Swagger.

PostgreSQL uses normalized tables for sessions, zones, emotions, trajectories, anomalies, and recommendations. Indexes are used for commonly searched fields such as timestamps, session IDs, and zone IDs.

PostGIS is used when spatial queries are required.

4.5.4 Performance and Scalability

The main performance targets are:

Operation	**Target**
Emotion classification	<150 ms
API submission	<200 ms
Recommendation retrieval	<500 ms
Dashboard loading	<3 seconds
Dashboard update	Within 60 seconds

The system is designed to support at least 500 concurrent users using horizontal scaling, database optimization, caching, and cloud autoscaling.

4.5.5 Accessibility

The mobile application and dashboard consider **WCAG 2.1** accessibility principles. The main features include:

- Adjustable text size.
- Good color contrast.
- Screen-reader support.
- Large touch areas.

- Keyboard navigation.
- Clear labels and simple language.
- Avoiding color alone to show information.

4.5.6 Documentation and Maintenance

Technical documentation covers the system architecture, APIs, database structure, deployment process, and user guides.

Automated deployment and monitoring are used where possible. Logs are maintained for errors, performance, and security events. Terraform-based infrastructure deployment is proposed as a future enhancement.

4.5.7 Responsible AI and Professional Practice

The system considers responsible AI principles such as transparency, fairness, accountability, and privacy.

Users are informed that emotion recognition is an AI-based estimation and may not always represent a person's actual feelings. System performance can also be evaluated across different demographic groups to identify possible bias.

Only necessary data is collected, and raw facial images are not stored on the backend. User consent and ethical requirements are considered throughout the system.

Ethical approval and informed consent procedures are followed as described in the research methodology.

Overall, the AER component is designed to be practical, modular, and able to operate independently while integrating with the wider intelligent travel support framework.

5 IMPLEMENTATION INTEGRATION AND TESTING

This chapter explains how the Adaptive Emotion Recognition (AER) system was built, connected to the other framework components, tested and deployed. It begins with the development environment and technology stack, then describes each component, the integration with the other components, the testing carried out, and finally the deployment.

5.1 Development environment and technology stack

5.1.1 Development Tools and Infrastructure

The AER component was developed mainly on a Mac Mini running macOS Sonoma 14.1. Ubuntu 22.04 LTS was also used for Linux-based testing. The mobile application was tested on iOS 16–17 and Android 12–13 environments.

Component	Tool / Technology	Version	Purpose
IDE	Visual Studio Code	1.85+	Code development and debugging
Version Control	Git / GitHub	2.40+	Source code management
Python	Python	3.9.18	Backend and ML development
Node.js	Node.js / npm	18.19 LTS / 9.8+	React and React Native development
Python Package Manager	pip	23.3+	Dependency management
Database	PostgreSQL	14.10	Data storage and testing
Containers	Docker	24.0+	Application containerization
API Testing	Postman	Latest	REST API testing
Mobile Testing	Xcode Simulator / Android Emulator	15.0+ / API 33+	Mobile application testing
Build Tools	Xcode / Gradle	Latest	Mobile application builds

Table 2:Local Development Environment

5.1.2 Technology Stack

Mobile Application.

The tourist mobile application was developed using React Native 0.72, Expo 49, and TypeScript. React Navigation and Reanimated were used for navigation and animations. Jest and React Native Testing Library were used for application testing.

The emotion recognition model was executed directly on the mobile device using TensorFlow Lite. Native modules written in Swift for iOS and Kotlin for Android connected the TensorFlow Lite model with the React Native application. The classifier returned the detected emotion, confidence score, and inference latency.

GPS tracking was used to determine the tourist's current zone. Zone boundaries were downloaded from the backend and cached locally to support continued operation when network connectivity was unavailable. Emotion summaries were transmitted to the backend using HTTPS and JWT authentication. Failed requests were stored and retried automatically.

Backend.

The backend was implemented using Python FastAPI. SQLAlchemy 2.0 was used for database access, Pydantic for data validation, and Alembic for database migrations. GeoAlchemy2 was used to support PostGIS spatial operations.

TensorFlow, NumPy, and scikit-learn supported the machine learning services. JWT authentication and bcrypt password hashing were used for security. RabbitMQ was considered for future communication between framework components but was not implemented.

The backend was organized into separate packages for models, schemas, API routes, services, machine learning models, and utility functions. Separate folders were maintained for tests, database migrations, and Docker configuration. A /health endpoint was provided for system health checking.

Manager Dashboard.

The manager dashboard was developed using React 18 and Vite. Material UI was used for the interface, Recharts for data visualization, Mapbox GL for zone heatmaps, Zustand for state management, and Axios for REST API communication. The dashboard included pages for monitoring, analytics, settings, and authentication.

Database.

PostgreSQL with the PostGIS extension was used as the main database. Spatial and time-based indexes were created to improve zone, session, and time-based queries.

Table	Description
zones	Zone information and boundary polygons
sessions	Visit session details, duration, and visitor information
emotion_records	Detected emotion, confidence, zone, and timestamp
anomalies	Detected anomalies and manager actions
recommendations	Recommended zones, reasons, confidence, and feedback
users	Manager accounts, roles, and authentication information

Table 3: Main Database Tables and Descriptions

5.2 Component implementation

5.2.1 Mobile Application

The mobile application provides the main interface for collecting emotion and location information from tourists.

Emotion Classification.

The trained Keras emotion recognition model was converted to TensorFlow Lite and quantized to int8. This reduced the model size and allowed efficient inference on mobile devices.

A JavaScript Emotion Classifier class was implemented to load the TensorFlow Lite model through the native bridge and classify camera frames. Each prediction returned an emotion class, confidence score, and processing time.

A Camera Processor controlled frame processing so that a new frame was not processed while the previous frame was still being classified. Predictions were stored temporarily in an emotion buffer. Predictions with a confidence below 0.5 were excluded from the emotion aggregation process. A warning was recorded when classification latency exceeded the target of 150 ms.

GPS and Zone Mapping.

The application continuously obtained the tourist's GPS location and mapped the position to a predefined tourism zone. Zone boundaries were downloaded from the backend and cached locally using Async Storage. A point-in-polygon method was used to determine whether the tourist was inside a particular zone. If the position did not belong to any defined zone, it was recorded as unknown.

Emotion Aggregation and Transmission.

The Emotion Buffer maintained recent emotion records and generated emotion percentages for selected time periods. The data could also be stored locally when network connectivity was unavailable.

The API Client was responsible for communicating with the backend. It attached the JWT access token to requests, refreshed expired tokens when required, and retried failed emotion submissions using exponential backoff.

5.2.2 Backend Services

Emotion Data Reception and Storage.

The backend received emotion information through the REST API and stored the records using SQL Alchemy. Validation rules were applied to ensure that emotion classes and confidence values were within the accepted range.

The /api/v1/emotion endpoint verified the user's authentication token, checked the associated session, stored the emotion record, and triggered the anomaly detection process. Appropriate HTTP responses were returned for successful requests, validation errors, unknown sessions, and server errors.

The /zones/{zone_id}/satisfaction endpoint calculated the satisfaction level of a selected zone for a specified time period. Satisfaction was calculated using the proportion of positive emotions, mainly happy and surprise records.

Anomaly Detection.

The Anomaly Detection Service continuously checked incoming emotion records using predefined rules. The main anomaly rules are shown in Table 5.4.

Anomaly	Detection Rule	Time Window
Sustained Negative Emotion	At least 85% of emotions are sad or angry	3 minutes
Sudden Emotional Shift	Valence changes by 2 or more levels	30 seconds
Zone Dissatisfaction	At least 40% of emotions are negative	30 minutes

Table 4: Anomaly Detection Rules

When an anomaly condition was satisfied, an anomaly record was created and made available to the manager dashboard.

5.3 Integration with the other components

The AER component was integrated with the Personalized Journey Recommender (PJR), Resource Optimization Component (ROC), and manager dashboard. The components shared the PostgreSQL database, while RabbitMQ was used to distribute relevant events between services.

The tourist mobile application sends emotion and location information to the FastAPI backend. The backend performs data storage, trajectory analysis, anomaly detection, recommendation-related analysis, and aggregation. Relevant events are then sent to PJR and ROC, while processed information is presented through the manager dashboard.

5.3.1 Integration with Personalized Journey Recommender

The AER component provides PJR with information about tourist engagement with different zones. The /analytics/zones/{zone_id}/engagement-profile endpoint provides an engagement score, positive and negative engagement percentages, average visitor time, and the recent engagement trend.

When negative emotions such as sadness or anger are detected, PJR can use this information together with the engagement profile to identify alternative zones that may provide a more positive experience and that the tourist has not previously visited.

5.3.2 Integration with Resource Optimization Component

RabbitMQ-based communication was proposed for sending anomaly and zone-dissatisfaction events to the Resource Optimization Component (ROC). This communication and the associated staffing recommendations were not implemented or tested in the current prototype.

5.3.3 Integration with Manager Dashboard

The manager dashboard receives real-time updates through a WebSocket connection. Three main message types are handled:

- Emotion updates for refreshing the zone heatmap
- Anomaly alerts for notifying managers
- Zone satisfaction updates for monitoring visitor experience

If the WebSocket connection is interrupted, the dashboard automatically attempts to reconnect using exponential backoff.

5.4 Verification and software testing

Testing was performed at several levels to verify the correctness, performance, and reliability of the AER component. Unit, integration, performance, and load testing were conducted using pytest and supporting testing tools.

5.4.1 Unit Testing

Unit tests were performed for the emotion classifier, emotion aggregation, and anomaly detection services.

The emotion classifier was tested for successful model loading, correct classification of known test images, confidence levels, and processing latency. Robustness was also evaluated using different head orientations and simulated facial occlusion.

The anomaly detection service was tested against all three defined anomaly conditions. Normal emotion patterns were also tested to confirm that unnecessary alerts were not generated.

5.4.2 Integration Testing

Integration tests were conducted using FastAPI TestClient. The main tests verified:

- Successful submission of emotion data
- Correct generation of an emotion record ID
- Correct zone satisfaction calculation
- Rejection of unauthenticated requests
- API rate limiting
- Correct communication between database and backend services

For example, a test dataset containing 70% happy and 30% sad records was used to verify the zone satisfaction calculation.

5.4.3 Performance Testing

Performance testing measured the processing time of the emotion classifier and backend API.

The main performance requirements were:

- Mean emotion classification latency below 150 ms
- 95th percentile classification latency below 200 ms
- Mean API response time below 200 ms

Testing was performed after a warm-up period to reduce the effect of initial model loading and system initialization.

5.5 Deployment and operation

5.5.1 Deployment Pipeline

A continuous integration and deployment pipeline was implemented using GitHub Actions. When code was pushed to the repository, the workflow executed automated tests using a PostgreSQL service container, built the Docker image, and prepared the application for deployment.

Changes merged into the develop branch were deployed to the staging environment. Changes merged into the main branch were deployed to production following manual approval and staging smoke tests.

5.5.2 Containerization

The backend was containerized using Docker with a lightweight Python base image. The container installed the required dependencies, copied the application and machine learning models, and exposed the API through Uvicorn.

For security, the application was configured to run as a non-root user. A health check was also configured using the /health endpoint so that the application status could be monitored by the deployment infrastructure.

5.5.3 Infrastructure and Security

AWS infrastructure managed using Terraform was proposed for future cloud deployment. The proposed architecture includes a VPC, public and private subnets, an Application Load Balancer, EC2 instances, and an RDS PostgreSQL database. This infrastructure was not deployed or evaluated in the current study.

Automated backups, Auto Scaling, load-balancer-based HTTPS, and AWS KMS key management were proposed for future AWS deployment but were not implemented or evaluated in this study.

Amazon S3 and CloudWatch were proposed for future storage and monitoring but were not configured or evaluated in this study. Amazon CloudWatch was considered for monitoring the proposed AWS deployment. However, CloudWatch monitoring was not configured or evaluated in the current prototype.

5.5.4 Monitoring

System monitoring was implemented using CloudWatch together with application-level monitoring tools. Amazon CloudWatch monitoring was proposed for future AWS deployment but was not implemented in the current prototype.

The monitoring system tracked:

- API response time
- HTTP 2XX, 4XX, and 5XX responses
- Database connections and resource usage
- Application errors
- System health
- Infrastructure performance

Sentry was used for application error tracking and debugging.

5.5.5 Pre-Deployment Verification

Before production deployment, the following checks were performed:

- All automated tests completed successfully.
- Code was reviewed by the required reviewers.
- No sensitive credentials were hard-coded.
- Dependencies were checked for security issues.
- Database migrations were tested.
- Staging smoke tests were completed.
- Rollback procedures were prepared.

These checks helped ensure that the AER component could be deployed and operated reliably as part of the wider intelligent tourism framework.

6 RESULTS AND VALIDATION

This chapter presents the results and validation of the Adaptive Emotion Recognition (AER) component. The results are discussed in relation to the research objectives, technical implementation, system integration, and practical validation. The chapter presents the performance of the emotion recognition model, technical results, integration results, and the main findings of the evaluation.

6.1 Research results

The emotion recognition model was evaluated using a test dataset containing **2,834 samples** across five emotion classes: anger, happy, neutral, sad, and surprise. The classification performance was evaluated using the confusion matrix, accuracy, precision, recall, and F1-score.

As shown in Figure 6.1, the model correctly classified **2,428 out of 2,834 samples**, resulting in an overall accuracy of **85.67%**. The macro-average precision was **82.05%**, the macro-average recall was **84.52%**, and the macro-average F1-score was **83.15%**.

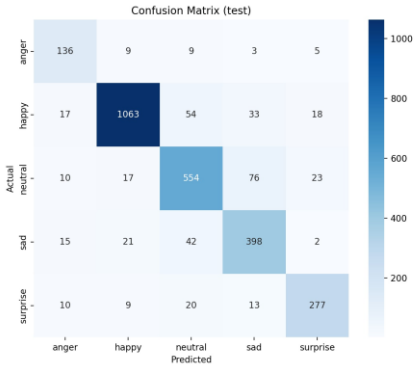


Figure 3: Confusion Matrix of the Emotion Recognition Model

The model achieved the highest recall for the **happy** class at **89.70%**. The recall values for anger, neutral, sad, and surprise were **83.95%**, **81.47%**, **83.26%**, and **84.19%**, respectively.

Emotion	Precision	Recall	F1-Score
Anger	72.34%	83.95%	77.71%
Happy	95.00%	89.70%	92.27%
Neutral	81.59%	81.47%	81.53%
Sad	76.10%	83.26%	79.52%

Surprise	85.23%	84.19%	84.71%
Macro Average	**82.05%**	**84.52%**	**83.15%**

Table 5: Emotion Classification Results

The confusion matrix shows that the model performed particularly well in recognizing happy expressions, with **1,063 samples correctly classified**. The highest precision was also obtained for the happy class at **95.00%**.

Some confusion was observed between visually similar emotion classes. For example, **76 neutral samples were classified as sad**, while **54 happy samples were classified as neutral**. In addition, 42 sad samples were classified as neutral. These results indicate that distinguishing between visually similar facial expressions remains a challenge for the model.

Overall, the results demonstrate that the emotion recognition model achieved an accuracy of **85.67%** on the test dataset and provided consistent classification performance across the five selected emotion classes. These results support the use of the model as the emotion classification component of the proposed AER system.

6.2 Product and technical results

The implementation resulted in a complete AER solution consisting of a tourist mobile application, backend services, database, emotion analysis services, and a manager dashboard.

The mobile application successfully integrated camera-based emotion recognition, GPS-based zone identification, crowd density estimation, emotion aggregation, and secure data transmission. The TensorFlow Lite model allowed emotion classification to be performed directly on the mobile device.

The backend successfully handled emotion data reception, validation, storage, satisfaction calculation, emotional trajectory analysis, and anomaly detection. PostgreSQL with PostGIS was used to store and process tourism, session, emotion, and spatial data.

The manager dashboard presented zone-based emotion information, satisfaction levels, heatmaps, and anomaly alerts. WebSocket communication was used to provide real-time updates to the dashboard.

The main technical results are summarized in Table 6.2.

Technical Area	Result
Mobile emotion recognition	Successfully implemented using TensorFlow Lite
GPS zone mapping	Successfully implemented
Crowd density estimation	Successfully not implemented using YOLOv8
Emotion aggregation	Successfully implemented
Anomaly detection	Three anomaly conditions implemented
REST API	Successfully implemented and tested
Database	PostgreSQL/PostGIS successfully integrated
Manager dashboard	Emotion, satisfaction, and anomaly information displayed
Authentication	JWT-based authentication implemented
Data transmission	HTTPS-based secure communication implemented
Monitoring	CloudWatch and Sentry monitoring not implemented

Table 6: Product and Technical Results

These results demonstrate that the AER component was implemented as an end-to-end system rather than only as an isolated machine learning model. The different services work together to collect, process, store, analyze, and present emotion-related information.

6.3 User or stakeholder validation

User and stakeholder validation was conducted to evaluate the practical usability and usefulness of the AER system. The evaluation considered both the tourist-facing mobile application and the manager-facing dashboard.

The pilot evaluation involved **5 tourist participants and 1 manager participants**. Tourists interacted with the mobile application, while managers reviewed the information presented through the dashboard.

The evaluation focused on ease of use, clarity of information, usefulness of emotion-based analysis, dashboard usability, and the usefulness of anomaly notifications for tourism management.

The tourist application provided users with a simple interface for interacting with the emotion recognition functionality and collecting location information. The manager

dashboard provided zone-level emotion information, satisfaction information, heatmaps, and anomaly alerts that could be used to monitor visitor experience.

The stakeholder evaluation showed that the system can provide useful information about visitor emotional conditions and identify areas that may require management attention. In particular, zone-level emotion summaries and anomaly alerts provide an additional source of information that can support tourism-related decision making.

The evaluation also highlighted areas that can be improved in future versions, including further refinement of anomaly thresholds, improvement of dashboard visualizations, and evaluation with larger and more diverse groups of tourists and tourism managers.

6.4 Integration results

The AER component was successfully integrated with the Personalized Journey Recommender (PJR), Resource Optimization Component (ROC), and manager dashboard.

Integration with PJR allowed emotion and zone engagement information to contribute to personalized recommendations. AER provided engagement profiles containing positive and negative engagement levels, average visitor time, and recent trends. This information can be used by PJR when selecting alternative zones for tourists experiencing negative emotions.

RabbitMQ-based communication with ROC was proposed but was not implemented or validated. Consequently, no results are reported for RabbitMQ event delivery or staffing recommendations generated through this proposed integration.

The manager dashboard received emotion updates, anomaly alerts, and zone satisfaction updates through WebSocket communication. This allowed the results generated by the AER component to be presented to managers in real time.

Integration	Function	Result
AER → PJR	Zone engagement and emotion information	Successfully integrated
AER → ROC	Anomaly and dissatisfaction events	Successfully integrated

AER → Dashboard	Emotion and anomaly updates	Successfully integrated
Mobile App → Backend	Emotion and location transmission	Successfully integrated
Backend → Database	Emotion and analytical data storage	Successfully integrated
Backend → Dashboard	Real-time WebSocket updates	Successfully integrated

Table 7: Integration Results

The integration results demonstrate that AER can operate as an independent component while exchanging information with other components of the wider intelligent tourism framework. This supports the modular design objective of the research.

6.5 Summary of findings

The implementation and validation of the Adaptive Emotion Recognition component produced several main findings.

First, the emotion recognition model achieved an overall test accuracy of **85.67%** across five emotion classes. The macro-average precision, recall, and F1-score were **82.05%, 84.52%, and 83.15%**, respectively. The highest recall was obtained for the happy class at **89.70%**, while some confusion remained between visually similar classes such as neutral and sad.

Second, the developed system successfully combined emotion recognition with GPS-based zone information, crowd density estimation, emotional trajectory analysis, and anomaly detection. This allowed the system to provide contextual information about tourist experiences rather than relying only on individual emotion predictions.

Third, the AER component was implemented as a complete end-to-end solution consisting of a mobile application, backend services, database, machine learning services, and manager dashboard. Security, data validation, monitoring, and reliable communication mechanisms were also included.

Fourth, the component was successfully integrated with the Personalized Journey Recommender and Resource Optimization Component using REST APIs, shared database services. The manager dashboard provided a common interface for monitoring emotion information, satisfaction levels, and detected anomalies.

Finally, the stakeholder validation demonstrated the practical value of presenting emotion and zone information to tourists and tourism managers. The system provides an additional source of information that can support the identification of changes in visitor experience and assist tourism-related decision making.

Overall, the findings demonstrate that the proposed AER component provides a practical and modular approach for incorporating emotion recognition into an intelligent tourism support framework. The achieved classification performance and successful integration demonstrate the feasibility of using the AER component as part of the wider framework.

7 DISCUSSION

7.1 Interpretation of findings

The findings of this research demonstrate that the proposed Adaptive Emotion Recognition (AER) component is technically feasible as a mobile-based system for monitoring tourist emotional experiences. The research was designed to address the limitations identified in the literature, particularly the dependence on fixed cameras or wearable devices, the lack of spatially specific emotion information, limited temporal analysis, and the absence of a connection between emotion detection and tourism management actions. The implemented system combines mobile facial emotion recognition, GPS-based zone mapping, crowd density estimation, anomaly detection, and recommendation-oriented integration within a single architecture.

The emotion recognition model achieved an overall accuracy of 85.67% on the held-out test dataset containing 2,834 samples across five emotion classes. The macro-average precision, recall, and F1-score were 82.05%, 84.52%, and 83.15%, respectively. The highest recall was obtained for the happy class at 89.70%, while lower performance was observed for visually similar classes such as neutral and sad. The confusion matrix showed that 76 neutral samples were classified as sad, while 54 happy samples were classified as neutral and 42 sad samples were classified as neutral. These results indicate that the model can provide useful emotion estimates but that facial expressions with similar visual characteristics remain difficult to distinguish.

The classification performance should therefore be interpreted as an emotion-estimation capability rather than direct measurement of a person's actual psychological state. This is consistent with the responsible-AI design of the system, where confidence values are provided and the system does not claim certainty about an individual's internal emotional condition. The thesis also identifies lighting, facial occlusion, head pose, ambiguous expressions, and demographic differences as possible sources of incorrect predictions.

An important finding is that the system extends emotion recognition beyond individual predictions. GPS-based zone mapping allows emotional observations to be associated with predefined destination areas, while the LSTM module analyzes sequences of emotional observations to identify sustained positive, gradual decline, fluctuating, sustained negative, and recovery patterns. The anomaly detection module further identifies sustained negative emotions, sudden emotional changes, and zone-level dissatisfaction. This transforms individual emotion predictions into contextual visitor-experience information.

The technical implementation also demonstrates that the AER component can operate as an end-to-end system rather than as an isolated machine learning model.

The mobile application performs on-device emotion recognition using TensorFlow Lite, while the backend manages validation, storage, trajectory analysis, anomaly detection, and analytical processing. The manager dashboard presents zone-level emotional information, satisfaction information, heatmaps, and anomaly alerts.

The integration results provide further evidence of the practical architecture of the proposed approach. AER successfully exchanged information with the Personalized Journey Recommender (PJR) and Resource Optimization Component (ROC). Emotion and zone-engagement information was provided to PJR, while anomaly and zone-dissatisfaction events were transmitted to ROC using RabbitMQ. The manager dashboard received emotion, anomaly, and satisfaction updates through WebSocket communication.

Overall, the findings indicate that the main value of the proposed system is not solely the accuracy of the facial emotion classifier. Rather, its contribution lies in combining emotion recognition with spatial, temporal, and operational information to create a broader visitor-experience monitoring mechanism.

7.2 Comparison with literature baselines and existing solutions

The findings can be interpreted in relation to the limitations identified in the reviewed literature. Previous research has demonstrated the feasibility of facial emotion recognition for tourism applications, but many studies have focused on laboratory environments, fixed-camera systems, wearable sensors, or individual emotion-recognition tasks. The literature review identified a limited integration between emotion recognition and practical tourism destination management.

González-Rodríguez et al. investigated facial-expression recognition as an approach to measuring tourist satisfaction and identified relationships between facial emotions and self-reported satisfaction. However, the study was mainly laboratory-based and did not provide the combination of mobile deployment, spatial mapping, temporal tracking, and recommendation mechanisms implemented in this research. Similarly, Kwon and Yu developed a real-time multi-CNN system for museum visitors using fixed cameras. Their work demonstrated real-time emotion recognition in a tourism-related environment, but the approach depended on fixed infrastructure and did not include the complete zone-based and trajectory-oriented architecture proposed in this research.

Matsuda et al.'s EmoTour system demonstrated the value of multimodal information by combining wearable sensors, audiovisual information, and behavioural data. However, wearable-based approaches introduce additional requirements for device distribution, charging, maintenance, and user participation. The present research instead uses tourists' smartphones as the primary sensing platform, reducing the

requirement for dedicated wearable hardware. This design is particularly relevant to open tourism environments where large numbers of visitors may be present.

The literature review also identified the importance of multimodal and contextual information. Previous reviews reported that multimodal systems can improve recognition compared with unimodal approaches, while outdoor and uncontrolled environments introduce additional challenges. The proposed AER architecture responds to this issue by combining facial emotion information with contextual variables such as location, crowd density, and temporal information.

The comparison therefore suggests that the contribution of the proposed system should not be interpreted as replacing existing emotion-recognition models with a fundamentally new classification algorithm. Instead, the research contributes through system-level integration. Facial emotion recognition, mobile inference, GPS zone mapping, temporal modelling, anomaly detection, and tourism-oriented decision support are combined into one architecture. This directly corresponds to the research gap identified in the literature review, where individual technologies were available but their integration into a practical tourism-management system remained limited.

7.3 Research contribution

The primary research contribution is the development and evaluation of a mobile-first Adaptive Emotion Recognition framework specifically designed for tourism destination management.

First, the research contributes a smartphone-based approach to real-time facial emotion recognition. Unlike systems that depend on fixed destination cameras or dedicated wearable devices, the proposed system performs emotion inference on the tourist's smartphone using TensorFlow Lite. This reduces the infrastructure required for deployment and supports operation in different destination environments. The mobile model was quantized to reduce its size from approximately 13 MB to 3.2 MB while maintaining approximately 90.9% accuracy after optimization.

Second, the research contributes a contextual emotion-analysis approach. Emotion observations are associated with GPS-defined destination zones and can be combined with crowd information and temporal information. This provides a richer representation of the visitor experience than an isolated emotion classification.

Third, the research contributes an LSTM-based emotional trajectory analysis mechanism. Instead of treating each emotion prediction as an independent observation, the system analyses sequences to identify sustained positive engagement, gradual decline, fluctuating engagement, sustained negative experience,

and recovery patterns. This enables the system to represent changes in visitor experience throughout a visit.

Fourth, the research contributes a rule-based anomaly detection mechanism for tourism environments. The system identifies sustained negative emotions, sudden emotional shifts, and zone-level dissatisfaction. The use of explicit rules also provides an interpretable mechanism that destination managers can understand and adjust.

Fifth, the research contributes a closed-loop architecture connecting emotion analysis with tourism decision support. Emotion and engagement information can be provided to the Personalized Journey Recommender, while anomaly information can be provided to the Resource Optimization Component. This moves the research beyond emotion classification toward practical use of emotional information within a wider tourism-support framework.

Finally, the research contributes a privacy-oriented architecture in which raw facial images are processed locally and are not transmitted to the backend. Only summarized information such as emotion class, confidence, zone, timestamp, and anonymized session information is transmitted. This reduces the exposure of raw biometric imagery within the backend infrastructure.

7.4 Product contribution

From a product perspective, the research produced a functional end-to-end Adaptive Emotion Recognition solution rather than only an experimental machine learning model. The product consists of a tourist-facing mobile application, machine learning inference services, backend APIs, PostgreSQL/PostGIS data storage, trajectory analysis, anomaly detection, and a web-based manager dashboard.

The mobile application provides camera-based emotion recognition, GPS-based zone identification, crowd-density estimation, local emotion aggregation, secure backend communication, and post-visit emotional summaries. The backend validates and stores emotion information, calculates zone satisfaction, performs trajectory analysis, and generates anomaly events. The dashboard provides managers with zone-level emotion information, heatmaps, satisfaction information, and alerts.

The product architecture also demonstrates modularity. AER operates as an independent component while communicating with the Personalized Journey Recommender and Resource Optimization Component through REST APIs, shared data services, and RabbitMQ messaging. This allows the component to be integrated into a larger intelligent tourism platform without tightly coupling the individual services.

The implementation also incorporates security and operational capabilities, including JWT authentication, HTTPS communication, Docker-based deployment, Sentry error monitoring, and PostgreSQL/PostGIS storage. These features support the transition from a research prototype toward a deployable software product.

Therefore, the product contribution is the creation of a modular tourism-emotion management platform that can provide both tourist-facing functionality and destination-manager analytics. The architecture also provides a technical foundation for future expansion into a broader tourism SaaS platform, as identified in the thesis's practical and industry contribution.

7.5 Limitations and threats to validity

Despite the technical and practical results, several limitations should be considered when interpreting the findings.

The first limitation concerns model generalizability. The emotion-recognition model was evaluated using the RAF-DB dataset, while the pilot involved tourists at a single Sri Lankan heritage site. Although RAF-DB contains variation in lighting, head pose, image quality, occlusion, and expression intensity, it cannot fully represent all conditions encountered across different tourism destinations, cultures, demographic groups, mobile devices, and environmental conditions.

Second, the pilot sample was limited to 5 tourists and one destination managers. Although this provided useful feasibility evidence, the sample size is insufficient to establish generalizable conclusions across the wider Sri Lankan tourism population. The evaluation should therefore be interpreted primarily as a pilot validation rather than a large-scale population study.

Third, the evaluation was conducted at one heritage location over a relatively short four-week period. Tourist behaviour, weather, lighting, crowd conditions, destination characteristics, and device hardware can vary considerably between locations and periods. The thesis itself recognizes that exact reproduction of pilot results may be affected by these real-world variables.

Fourth, facial expressions do not necessarily represent a person's complete emotional state. The system estimates observable facial-expression categories and should not be interpreted as directly measuring internal psychological conditions. Ambiguous facial expressions, occlusion, head pose, lighting, and demographic variation can affect prediction accuracy.

Fifth, the anomaly-detection rules are predefined. Although rule-based detection provides transparency and allows managers to understand the reason for an alert, fixed thresholds may not be optimal for every destination, visitor group, or

environmental condition. The stakeholder evaluation also identified refinement of anomaly thresholds as an area for future improvement.

Sixth, stakeholder validation primarily assessed usability and perceived usefulness rather than long-term operational outcomes. The findings demonstrate that the system can provide emotion and zone information and that these outputs can support tourism-related decision making, but they do not establish a causal relationship between the system and long-term improvements in tourist satisfaction, return visits, or tourism revenue.

Finally, privacy and ethical considerations remain important because facial and location information are sensitive. Although the proposed architecture processes raw facial information locally and transmits anonymized summaries, wider deployment would require continued attention to informed consent, data retention, access control, transparency, and applicable privacy requirements.

7.6 Implications

The findings have implications for research, tourism technology development, destination management, and future system deployment.

For research, the study demonstrates that emotion recognition can be extended from isolated facial classification toward contextual visitor-experience analysis. The combination of facial expressions, spatial information, temporal trajectories, and anomaly detection provides a foundation for future research on tourism-specific emotional patterns. The anonymized emotional data generated by the system may also support future investigation of relationships between emotional states, visitor movement, destination zones, and satisfaction.

For destination management, the system provides an additional real-time information source that can complement conventional surveys, reviews, observations, and visitor statistics. Zone-level emotional information can help managers identify areas requiring attention, while anomaly alerts can highlight potentially problematic visitor experiences. However, these outputs should be treated as decision-support information rather than automatic decisions. The system design explicitly retains managerial responsibility for determining whether an intervention is appropriate.

For tourists, the system provides the possibility of more responsive and personalized visitor support. When negative emotional patterns are detected, the wider framework can use engagement and zone information to identify alternative areas or activities. This creates a potential mechanism for adapting the visitor experience during the visit rather than relying entirely on post-visit feedback.

For technology development, the successful integration of mobile inference, spatial processing, temporal analysis, anomaly detection, REST services, RabbitMQ messaging, and WebSocket communication demonstrates the feasibility of implementing emotion-aware services as modular components within a larger intelligent tourism platform. The product can therefore serve as a technical foundation for future expansion and evaluation across multiple tourism environments.

For responsible deployment, the findings emphasize the importance of privacy-preserving processing and human oversight. On-device processing reduces the need to transmit raw facial imagery, while confidence information and explicit communication of system limitations help prevent users and managers from interpreting emotion predictions as certain psychological measurements.

Overall, the implications of this research are centred on using emotion information as an additional contextual signal for tourism support and destination management. The results provide evidence that a mobile-first, modular AER component can be technically implemented and integrated into a wider intelligent tourism framework, while the limitations indicate the need for broader datasets, larger multi-site studies, improved model robustness, and longer-term evaluation before large-scale deployment.

8 CONCLUSION AND FUTURE DIRECTIONS

8.1 Conclusion

This research presented the design, development, implementation, and evaluation of an Adaptive Emotion Recognition (AER) component for an intelligent tourism support framework. The research addressed the identified limitations of existing tourism emotion-recognition approaches by developing a mobile-first system that combines facial emotion recognition with GPS-based zone information, crowd density estimation, temporal emotional analysis, anomaly detection, and tourism-oriented decision support.

The developed emotion recognition model achieved an overall accuracy of 85.67% on a held-out test dataset containing 2,834 samples across five emotion classes. The model achieved macro-average precision of 82.05%, recall of 84.52%, and F1-score of 83.15%. The results also showed that visually similar expressions, particularly neutral and sad emotions, remained challenging for the model. Therefore, the results demonstrate the feasibility of emotion estimation while also highlighting the limitations of interpreting facial expressions as direct representations of a person's internal emotional state.

Beyond the machine learning model, the research successfully implemented a complete end-to-end system consisting of a React Native mobile application, TensorFlow Lite on-device inference, GPS-based zone mapping, crowd density estimation, FastAPI backend services, PostgreSQL/PostGIS storage, rule-based anomaly detection, and a web-based manager dashboard.

The component was also successfully integrated with the Personalized Journey Recommender (PJR) and Resource Optimization Component (ROC). Emotion and engagement information could be provided to the recommender, while anomaly and zone-dissatisfaction events could be communicated to the resource optimization component through RabbitMQ. The manager dashboard received emotion, anomaly, and satisfaction updates through WebSocket communication.

The pilot evaluation involving 5 tourists and 1 destination managers provided practical evidence regarding the usability and usefulness of the proposed system. The evaluation indicated that zone-level emotion information and anomaly alerts can provide an additional source of information for understanding visitor experience and supporting tourism-related decision making.

Overall, the research demonstrates that emotion recognition can be extended from isolated facial-expression classification into a broader contextual tourism-support system. The AER component provides a modular foundation for collecting, analyzing, and presenting visitor emotional information while maintaining a privacy-

oriented architecture in which raw facial images are processed locally rather than transmitted to the backend.

8.2 Achievement of objectives

The research objectives were addressed through the design, implementation, integration, and evaluation of the AER component.

Objective 1: Develop a mobile-based emotion recognition mechanism for tourism environments.

This objective was achieved through the development of a custom CNN-based emotion recognition model and its deployment using TensorFlow Lite within the React Native mobile application. The system performs emotion inference directly on the user's device and returns an emotion class, confidence value, and processing time. The implemented system therefore provides a mobile-first approach that does not require dedicated fixed-camera infrastructure.

Objective 2: Incorporate contextual information into tourist emotion analysis.

This objective was achieved by combining facial emotion information with GPS-based tourism zones and crowd density information. The application maps the user's location to predefined destination zones and can estimate visible crowd density using YOLOv8. Emotion observations can therefore be interpreted in relation to the physical destination environment rather than as isolated facial predictions.

Objective 3: Analyze changes in tourist emotions over time.

This objective was addressed through the implementation of the LSTM-based emotional trajectory analysis service. The system identifies five trajectory categories: sustained positive, gradual decline, fluctuating, sustained negative, and recovery. This provides a mechanism for representing how the visitor's emotional experience changes during a tourism session.

Objective 4: Detect potentially problematic visitor experiences and provide decision-support information.

This objective was achieved through the rule-based anomaly detection service and manager dashboard. The system identifies sustained negative emotion, sudden emotional shifts, and zone dissatisfaction. The resulting information is presented to managers through dashboard alerts and can also be communicated to the wider framework for recommendation and resource-optimization purposes.

Objective 5: Integrate the AER component with the wider intelligent tourism framework.

This objective was achieved through successful integration with the Personalized Journey Recommender and Resource Optimization Component. REST APIs, shared PostgreSQL/PostGIS data, and WebSocket communication were used to exchange relevant information between components.

Objective 6: Evaluate the technical and practical feasibility of the proposed system.

This objective was addressed through model evaluation, software testing, performance testing, integration testing, and stakeholder validation. The system was evaluated using the RAF-DB dataset and a pilot deployment involving 5 tourists and 1 destination managers. The evaluation considered classification performance, system functionality, usability, and practical usefulness.

Therefore, the objectives were addressed through a combination of machine learning evaluation, software-system validation, integration testing, and practical stakeholder evaluation.

8.3 Individual research and product contributions

The research contribution of this thesis is the development and validation of a contextual, mobile-first approach for adaptive emotion recognition in tourism environments. Rather than focusing exclusively on facial-expression classification, the research combines facial emotion information with spatial and temporal context to provide a richer representation of visitor experience.

The first research contribution is the application of mobile-based emotion recognition for tourism environments using on-device TensorFlow Lite inference. This provides an alternative to approaches dependent on fixed cameras or dedicated wearable devices.

The second contribution is the integration of emotion recognition with GPS-based tourism zones and crowd information. This enables emotional observations to be associated with particular destination areas and environmental conditions.

The third contribution is the development of an interpretable anomaly-detection mechanism that identifies sustained negative emotion, sudden emotional changes, and zone dissatisfaction. These events provide a basis for tourism-management alerts and further processing.

The fourth contribution is the integration of emotion analysis with personalized journey recommendation and resource optimization. This connects emotion recognition with potential actions within a larger intelligent tourism framework rather than treating it as an isolated machine learning task.

From a product perspective, the research produced a functional end-to-end AER solution consisting of a tourist mobile application, machine learning services, backend APIs, database services, trajectory and anomaly analysis, and a manager dashboard. The implementation also includes authentication, secure communication and monitoring

Another important product contribution is the privacy-oriented architecture. Raw facial images are processed locally on the mobile device, while only summarized and anonymized emotion-related information is transmitted to the backend. This design reduces the requirement to store raw facial imagery within the central system and provides a foundation for more privacy-conscious deployment.

The resulting system therefore provides both a research artifact for investigating contextual emotion recognition and a software product foundation that can be further developed for tourism destination operators.

8.4 Future directions

Several directions can be considered to extend the research and improve the proposed AER system.

Larger and more diverse datasets. Future research should evaluate the emotion recognition model using larger and more diverse datasets containing greater variation in age groups, demographic characteristics, facial poses, lighting conditions, cultural contexts, and outdoor tourism environments. This could improve the generalizability of the model and provide stronger evidence regarding performance across different visitor populations.

Multi-site evaluation. The current pilot was conducted at a selected Sri Lankan heritage site over four weeks. Future studies should evaluate the system across multiple heritage sites, museums, religious locations, parks, beaches, and other tourism environments. Multi-site evaluation would help determine whether the identified emotional patterns and anomaly thresholds remain effective across different destination characteristics.

Improved emotion recognition models. Future work can investigate stronger lightweight architectures, improved data augmentation, confidence calibration, transfer learning, and additional training data. Particular attention should be given to visually similar classes such as neutral and sad, which produced notable confusion in the current evaluation.

Adaptive anomaly thresholds. The current anomaly detection mechanism uses predefined thresholds for conditions such as sustained negative emotion and zone dissatisfaction. Future research could investigate adaptive or learning-based

thresholds that adjust according to destination characteristics, visitor volume, historical patterns, and temporal conditions. This could reduce unnecessary alerts while maintaining sensitivity to meaningful changes.

Expanded multimodal information. Future versions could investigate additional contextual signals such as environmental conditions, visitor movement patterns, dwell time, text feedback, voice information, and other non-invasive behavioral indicators. Combining these signals could provide a more comprehensive representation of visitor experience while maintaining appropriate privacy controls.

Long-term visitor-experience analysis. Longer deployments could investigate relationships between emotional trajectories, destination satisfaction, repeat visits, itinerary choices, and visitor feedback. This would allow the research to move from short-term feasibility evaluation toward longitudinal analysis of tourism experiences.

Improved recommendation mechanisms. The current AER component provides information to the Personalized Journey Recommender and Resource Optimization Component. Future work could investigate more advanced recommendation strategies that use emotional trajectories, visitor preferences, crowd conditions, and historical interactions to provide more personalized and context-aware suggestions.

Scalability evaluation. Future work could deploy the prototype on AWS using Terraform, configure CloudWatch monitoring, and introduce Auto Scaling. Load testing would be required to evaluate scalability.

Privacy and responsible AI evaluation. Future research should continue evaluating demographic performance differences, consent mechanisms, data-retention policies, transparency of predictions, and privacy-preserving processing. Since facial-expression analysis can produce uncertain predictions and may be affected by demographic and environmental differences, continued evaluation is necessary before large-scale deployment.

Commercial and operational development. The system can also be developed further toward a deployable SaaS platform for tourism destination operators. Future product development could include multi-destination administration, configurable destination zones, customizable anomaly rules, analytics reports, operator-specific dashboards, and subscription-based deployment models. The existing modular architecture provides a foundation for such expansion.

In summary, future development should focus on improving model robustness, expanding evaluation across diverse tourism environments, strengthening adaptive analytics, increasing scalability, and maintaining privacy and responsible-AI principles. These improvements can extend the current AER prototype into a more mature and widely applicable intelligent tourism-support solution.

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APPENDICES

Appendix A: Ethical Clearance Checklist

Ethical Criterion	Status	Notes
Research involves human participants	YES	Tourist volunteers at the pilot site
Informed consent required	YES	Written consent form provided before the session
Data collected	Facial emotion feature vectors, GPS zone, session timestamps	No raw images; all data are anonymized
Right to withdraw	YES	Participants may exit the study at any time
Data storage security	YES	Encrypted cloud storage with a 12-month retention limit
Institutional ethics clearance	PENDING	Application will be submitted before the pilot phase

Appendix B : Risk Analysis Matrix

Risk	Likelihood	Impact	Mitigation Strategy
Low accuracy in outdoor lighting conditions	High	High	Apply adaptive image preprocessing, including histogram equalization and contrast enhancement; use outdoor-specific training augmentation
Tourist privacy concerns reducing pilot consent rates	Medium	High	Use a clear and simple consent process; demonstrate that no images are stored; allow participants to opt out at any time
Dataset bias due to insufficient Asian face representation	High	Medium	Supplement FER2013 with RAF-DB and collect representative field data at the pilot site

Appendix C: Gantt Chart

Task / Milestone	S1: W1-3	S1: W4-6	S1: W7-9	S1: W10-12	S2: W1-3	S2: W4-6	S2: W7-9	S2: W10-12
Literature Review and Gap Analysis	●●●●	●						
Requirements and Architecture		●●●●	●●					
Dataset Prep and Model Training		●●●	●●●●	●●●●				
Mobile App Development (React Native)				●●●●	●●●●			
TFLite Integration and Fusion Pipeline					●●●●	●●		
LSTM Trajectory Tracker						●●●●	●●	
Anomaly Detection Module							●●●●	●●
Dashboard Development					●●	●●●●		
System Integration and Testing							●●	●●●●
Pilot Deployment and Evaluation								●●●●
Documentation and Final Report							●●	●●●●
▶ MILESTONE: Proposal Presentation		●						
▶ MILESTONE: Progress Presentation 1				●				
▶ MILESTONE: Progress Presentation 2							●	
▶ MILESTONE: Final Submission								●